

Coalition Design under Heterogeneous AV Objectives: Regimes and Low-Regret Selection in Mixed-Autonomy Networks

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Abstract

When several autonomous-vehicle (AV) operators share a road network, should their routing be centralized, or should some independent decision centers be retained? We study this question when operators have heterogeneous management objectives, aggregate demand is fixed, and human-driven vehicles (HDVs) continue to reroute. Each coalition is an atomic splittable player with a convex management objective and member-level origin–destination obligations; HDVs form a nonatomic Wardrop population. On a fixed affine support, we project each coalition response onto its demand-feasible tangent and remove the directions that HDVs can reproduce. The resulting activated response signature is sufficient for aggregate equilibrium and minimal for the conditional response map. It yields a general-network merge-direction score: stronger coupling improves congestion exactly when the screened response has negative system-cost projection. A canonical bottleneck theorem then identifies three regimes. Coalition organization is neutral when HDVs fully screen the response; centralization is favored when it moves effective response toward the relevant boundary; and partial coordination can dominate both endpoints when the system-cost-minimizing response is interior. We evaluate local merge policies against complete four-company audit scans. In Sioux Falls, one- and two-step policies attain an optimal arrangement in all four tested states, whereas an always-grand rule loses up to 18.695 recovery percentage points. Across 480 certified profiles on independent Braess and directed-grid networks, one-step merging is exact in 29 of 32 states with mean regret 0.332 points, while two-step lookahead is exact in all 32. Both networks move from grand to partial coordination at high AV penetration under sufficiently strong governance, although the transition boundary differs. Adding an explicit governance burden generates grand, partial, and fragmented policy intervals. HHI remains descriptive because it does not identify objective composition, demand authority, or the response that survives HDV reallocation. The results concern conditional continuation design, not endogenous coalition formation or empirical calibration of operator objectives.

Keywords: mixed autonomy; coalition structure; heterogeneous AV management objectives; atomic splittable routing; Nash–Wardrop equilibrium; congestion; fleet organization

1 Introduction

The transition to mixed-autonomy traffic creates an organizational problem in addition to a control problem. Autonomous vehicles (AVs) will be dispatched by multiple firms, platforms, or routing authorities, while human-driven vehicles (HDVs) continue to respond to congestion through individual route choice. The firms need not value the same operational outcome: one may emphasize the travel time of its own fleet, another may internalize a larger share of congestion, and a third

may encode reliability, energy, or service obligations. The same set of firms can therefore be operated as independent routing centers, coordinated in partial coalitions, or placed under a common dispatcher. These arrangements are not equivalent when the aggregate demand and the physical road network are held fixed.

This paper studies the resulting design problem. Given a set of heterogeneous AV companies, when should their routing be centralized, when should partial independence be retained, and can a local rule select the organization without relying on a full ex post enumeration of partitions? The question is deliberately conditional. A partition and a governance rule are specified before routing begins; companies retain the OD obligations associated with the trips they have accepted; and the analysis compares the equilibria induced by alternative arrangements. We do not model whether firms merge or sign an agreement. The object of interest is the operational consequence of an arrangement that a platform, fleet manager, or public authority could choose among feasible alternatives.

The need for this distinction is practical as well as theoretical. Ownership concentration statistics such as the Herfindahl–Hirschman Index (HHI) summarize who controls how much mass and are used in market-screening contexts (?). They do not record what those firms optimize, which firms are grouped together, how many routing decision centers remain, which OD obligations each member must preserve, or which parts of a coalition response can be offset by HDV rerouting. Consequently, two organizations with the same company HHI, coalition size, and aggregate demand can induce different network flows. The conditional design problem can be written as

$$\Pi^*(\alpha, \theta, \gamma) \in \arg \min_{\Pi \in \mathfrak{P}(N)} J(x^*(\Pi; \alpha, \theta, \gamma)), \quad (1)$$

where Π is a feasible partition of the companies, γ is a governance intensity, α is AV penetration, θ collects objective types, and x^* is the mixed-autonomy equilibrium induced by that arrangement. The map is a design object, not an endogenous coalition-formation equilibrium.

We implement this design problem as a coalition-structured mixed Nash–Wardrop game. Each coalition is an atomic splittable player with a convex management objective, while HDVs form a nonatomic Wardrop population. All vehicles experience the same physical edge cost; heterogeneity enters through the objectives used to manage AV flows. The main treatment uses identity-preserving coordination: members may jointly choose routes, but each member must satisfy its own accepted OD demand. Operational pooling, which relaxes those member-level constraints, is analyzed separately because it changes feasible authority rather than merely changing objective aggregation. The formulation links atomic congestion games and Cournot–Wardrop interaction (????) with multiclass assignment (?) and mixed-autonomy routing (???).

The model makes the transmission mechanism explicit. Demand conservation restricts every coalition’s route response to an OD-feasible tangent. After this response is aggregated in edge space, HDVs can absorb any component that lies in their own Wardrop adjustment space. The remaining component is the activated coalition response. Projecting that response onto the gradient of system travel time produces a directional test for a proposed merge. The criterion distinguishes a screened regime, in which organization is locally neutral, from regimes in which stronger coupling raises or lowers congestion. In the canonical bottleneck, it also identifies an interior effective-response target at which partial coordination can dominate both grand and singleton organization.

Our first contribution is a unified representation of heterogeneous coalition control in mixed-autonomy routing. The model separates the physical cost function shared by all vehicles from the company management objective, the coalition governance matrix, the number of underlying companies n , and the number of active coalition decision centers $K = |\Pi|$. The convex quadratic family contains independent routing, partial cross-member internalization, and common-dispatcher

arrangements while preserving the OD obligations needed for a clean coalition comparison.

Our second contribution is a regime theory for coalition design. Starting from the bordered sensitivity systems used for equilibrium networks with polyhedral constraints (????), we derive the OD-restricted coalition response, aggregate it in edge space, and quotient it by the endogenous HDV Wardrop tangent. On a fixed affine support, the activated response signature is sufficient for aggregate equilibrium and minimal for the complete conditional response map. Its system-cost projection gives a general-network merge-direction score. The canonical bottleneck sharpens this score into a threshold theorem: complete HDV screening makes organization neutral, a boundary regime favors changes that reduce effective response, and an interior regime can favor partial coordination. Parametric equilibrium computation supplies the complementary support-traversal perspective (?).

Our third contribution is a certified, regret-based selection framework. We evaluate one- and two-step agglomerative policies against complete four-company scans, using enumeration only to audit the selected partitions. Both policies attain a state optimum in all four Sioux Falls conditions. On independent Braess and directed-grid networks, the one-step rule is exact in 29 of 32 states with mean regret 0.332 recovery points, while two-step lookahead is exact in all 32 states. Across the 480 certified cross-network profiles, grand coordination is preferred at lower penetration, whereas partial coordination emerges at high penetration under sufficiently strong governance on both topologies. An explicit governance-burden model produces calculable intervals in which grand, partial, or fragmented control is optimal. Composition sensitivity, company-count, spatial-exposure, and HHI diagnostics establish why the policy must use the response state rather than a mass statistic.

The numerical evidence is deliberately bounded. The three networks are computational benchmarks, not an empirical sample of mobility markets. The reported rankings hold within the declared objective family, identity-preserving governance rule, demand portfolios, and route sets; the affine regime results are not asserted as global BPR theorems. The two-step policy’s zero observed regret is a benchmark result rather than a universal guarantee. Pricing, passenger acceptance, empty-vehicle repositioning, platform profits, empirical calibration, and endogenous formation remain outside the paper’s scope.

The remainder of the paper follows this logic. Section 3 defines the physical network, company objectives, coalition governance, equilibrium, and performance measures. Section 4 establishes well-posedness, the activated response, the general merge score, and the canonical regimes. Section 5 interprets coalition composition, governance burden, HHI, and the boundary between objective coordination and pooling. Section 6 reports mechanism, regret, and cross-network evidence. Section 7 discusses implementation and scope, and Section 8 summarizes the bounded implications.

2 Positioning in the Literature

2.1 Multiple operators and mixed-autonomy networks

Research on automated mobility-on-demand and ride-sourcing has established that several operators can compete for demand while interacting through congestion. Existing formulations study operator competition and market decisions (??), jointly model ride-hailing competition with transit planning (?), or embed ride-sourcing services in urban traffic-network equilibrium (?). A related control literature studies the equilibrium and incentive consequences of routing AVs alongside individually routed vehicles (???). These strands motivate the distinction between positive-mass controlled flows and nonatomic HDVs, but they leave a different organizational question open: conditional on fixed demand, how should heterogeneous routing objectives be grouped into independent or coordinated decision centers? Operator count or market share alone does not answer

that question, because it does not specify objective composition, coalition governance, or member-level OD authority. The present paper isolates that operational layer while holding the physical road-cost function common across vehicle classes.

2.2 Atomic congestion games, collusion, and coalition structure

Atomic congestion and Nash–Wardrop models provide the natural language for positive-mass routing authorities whose choices affect the costs faced by other users (????). Multiclass assignment and strategic routing further show how equilibrium depends on the interaction between populations with different decision rules (??). Work on collusion and concentration in congestion games demonstrates that changing the strategic grouping of users can alter equilibrium efficiency (??). The gap relevant here is not the existence of coalition effects; it is their identification in a mixed-autonomy network with heterogeneous company objectives and fixed OD obligations. A coalition in our model is therefore more than a larger atomic mass. Its governance matrix determines which member effects are internalized, while the partition determines how many strategic response centers remain. This distinction allows the same company masses and coalition HHI to produce different continuation equilibria.

2.3 Equilibrium sensitivity and the transmission of organizational change

Sensitivity analysis for equilibrium network flows uses active supports, bordered systems, and derivatives restricted to polyhedral feasible sets (????). Recent work on atomic splittable games addresses the complementary computational problem of locating and traversing parametric equilibrium supports (?). We use these tools for their established purpose and do not present the bordered inverse as a new numerical primitive. The paper’s addition is to compose three response blocks that are usually kept separate: the OD-restricted response of each coalition, its edge-space aggregation, and the endogenous HDV Wardrop adjustment space. The resulting activated signature identifies the part of an organizational change that can reach aggregate congestion on a fixed support. This composition is also why scalar concentration summaries need not order network outcomes: the same coalition-side displacement can be fully screened in one HDV support and remain active in another.

2.4 Ownership, information, and interoperability

Ownership and information architecture are related institutional choices but are not interchangeable operational mechanisms. HHI is an ownership-concentration statistic used in merger screening (?); information-design studies show that changing what users know can itself alter network congestion and need not improve it monotonically (??). Transport policy and data-system work likewise treats interoperability as a coordination standard that can be implemented without common ownership (??). These sources motivate a separation that is central to our design interpretation. Identity-preserving coalition coordination changes the aggregation of management objectives at fixed member authority; operational pooling changes the feasible authority by relaxing member-level OD constraints; and information sharing is a possible microfoundation for better response estimation, not a guarantee of lower network travel time. The paper keeps these channels separate in both the model and the experiments.

2.5 Position of this paper

The literature supplies the ingredients for our analysis but does not connect them to a regime and selection theory for coalition organization. We combine atomic coalition control, mixed-autonomy equilibrium, and support-conditional sensitivity in a model with common physical costs and heterogeneous AV management objectives. The activated response yields a general-network merge score and a canonical threshold at which partial coordination can dominate both endpoints. Complete partition scans then serve as audit baselines for local policies with measurable regret, while governance burden produces explicit organizational intervals. The resulting claim is operational and bounded: coalition design matches a response state, whereas HHI and coalition size remain descriptors rather than sufficient equilibrium controls.

3 A Coalition-Structured Mixed Routing Game

This section defines the continuation game that is evaluated for each candidate coalition arrangement. The distinction between three objects is maintained throughout. The physical link cost is a property of the road and is common to all vehicles. A company's management objective describes how it values the routing of its own accepted AV requests. A coalition structure determines which companies choose routes jointly and which member-level obligations remain binding. Coalition effects therefore operate through decision rights and objective aggregation; they do not alter the road technology. The partition is an exogenous institutional treatment in this paper, so the model evaluates the congestion consequence of an arrangement rather than a firm's incentive to form it.

3.1 Network, routes, and fixed demand

Traffic uses a finite directed graph $G = (V, E)$ and a finite set W of origin–destination (OD) pairs. For each $w \in W$, total demand $d_w > 0$ is fixed and the admissible route set is $\mathcal{R}_w$. Let $\mathcal{R} = \bigcup_{w \in W} \mathcal{R}_w$, and let $\delta_{er} = 1$ if edge e belongs to route r . For any population, route flows f induce edge flow

$$x_e = \sum_{r \in \mathcal{R}} \delta_{er} f_r. \quad (2)$$

Every vehicle experiences the same physical link cost $c_e(x_e)$. This common-cost assumption is what makes AV governance a behavioral mechanism rather than a vehicle-class-specific road technology. The numerical model uses the Bureau of Public Roads function

$$c_e(x_e) = t_e^0 \left[1 + B \left(\frac{x_e}{\kappa_e} \right)^\eta \right], \quad B = 0.15, \quad \eta = 4, \quad (3)$$

where t_e^0 is free-flow time and κ_e is capacity. The theoretical results require only that c_e be continuous and nondecreasing; affine results use

$$c_e(x_e) = a_e + b_e x_e, \quad b_e \geq 0. \quad (4)$$

The experienced cost of route r is $C_r(x) = \sum_{e \in E} \delta_{er} c_e(x_e)$. The BPR specification is used for numerical continuation experiments; it is not used to claim the affine closed form below.

3.2 Companies, AV demand, and coalition partitions

Let $N = \{1, \dots, n\}$ be the underlying AV companies. Company i controls an accepted OD-demand vector $d^i = (d_w^i)_{w \in W}$ and mass $m_i = \sum_w d_w^i$. HDV demand is $d^H = (d_w^H)_{w \in W}$, and the realized

demand batch satisfies

$$d_w^H + \sum_{i \in N} d_w^i = d_w, \quad w \in W. \quad (5)$$

The routing stage is conditional on this accepted batch. Request acceptance, pricing, passenger platform choice, empty-vehicle repositioning, and inter-batch balancing are outside the model. Thus a company can change the path of an assigned request but cannot change its OD pair during the routing game. Holding the batch fixed is deliberate: it lets the comparisons identify routing and governance effects without conflating them with market-share or demand-formation effects.

A coalition structure is a partition $\Pi = \{C_1, \dots, C_K\}$ of N , where $K = |\Pi|$ is the number of independent routing decision centers. The number n of underlying companies and the number K of active coalitions are distinct: changing n changes the population of objectives, whereas changing K changes which objectives are jointly optimized. Company-level HHI is a function of the m_i ; coalition-level HHI is a function of the aggregate masses $m_C = \sum_{i \in C} m_i$. Neither index defines the coalition's management objective, and neither is assumed to order equilibria.

For company i , let y^i denote its route-flow vector and let Λ_i map routes to company edge flows. A coalition C may use the stacked member vector $y^C = (y^i)_{i \in C}$. Under the main identity-preserving governance rule, every member retains its own OD obligation:

$$\mathcal{Y}_C^{\text{id}} = \{y^C \geq 0 : \Gamma_C y^C = d^C\}, \quad \Gamma_C = \text{blkdiag}(\Gamma_i)_{i \in C}. \quad (6)$$

The coalition jointly chooses y^C but cannot reassign an accepted passenger obligation from one member to another. This is the main treatment because it changes the decision maker while holding ownership of requests fixed; the resulting comparison isolates objective aggregation and coordination from a change in service authority.

An operational-pooling governance rule instead preserves only the coalition total on each OD pair:

$$\mathcal{Y}_C^{\text{pool}} = \left\{ y^C \geq 0 : \sum_{i \in C} \sum_{r \in \mathcal{R}_w} y_r^i = \sum_{i \in C} d_w^i, \forall w \right\}. \quad (7)$$

Pooling is a separate intervention because it changes the feasible set, whereas identity-preserving coordination changes only the objective aggregation and cross-member internalization. The distinction matters empirically: a pooling result cannot be interpreted as evidence about coordination alone when pooling also permits one member to serve another member's accepted demand.

Proposition 1 (Pooling changes feasible authority). *For every coalition C , $\mathcal{Y}_C^{\text{id}} \subseteq \mathcal{Y}_C^{\text{pool}}$. The inclusion can be strict. If all members have identical admissible routes for every OD pair and the coalition objective depends only on aggregate coalition flow, the two sets induce the same aggregate route-flow set.*

Proof. Summing the member-specific equalities in (6) gives the pooled equalities in (7). Hence the identity-preserving set is contained in the pooling set. The inclusion is strict whenever a pooled solution allocates more than d_w^i of OD- w flow to member i . Conversely, if all members share the same route set, any pooled aggregate route vector $\bar{y}_w$ can be split as $y_w^i = (d_w^i/d_w^C)\bar{y}_w$, where $d_w^C = \sum_{i \in C} d_w^i$. This split satisfies every member equality and produces the same aggregate vector. If the objective depends only on that aggregate vector, the induced aggregate optimization problems coincide. $\square$

3.3 Company management objectives

Let $F_e^C = \Lambda_C y^C$ be coalition C 's edge flow and $x_e^{-C} = x_e - F_e^C$ be the flow controlled by other populations. The coalition management objective is

$$\Phi_C(y^C; y^{-C}, h) = \sum_{e \in E} \int_0^{F_e^C} c_e(x_e^{-C} + s) ds + \Psi_C(y^C). \quad (8)$$

The first term is the common physical routing component. It is written as an integral because, conditional on the other populations' flow, its derivative with respect to coalition flow is the experienced link cost $c_e(x_e)$. This gives the coalition the correct marginal road delay while preserving a convex objective whenever c_e is nondecreasing. The convex function Ψ_C is the company- and governance-specific management objective. It may represent own-fleet congestion internalization, energy, reliability, service commitments, route preferences, or cross-member coordination. These interpretations are alternative economic readings of the same reduced-form routing objective; the numerical design below uses the congestion-curvature component and does not claim to estimate each interpretation separately.

The route marginal field used by coalition C is

$$G_C(y) = \nabla_{y^C} \Phi_C = \Lambda_C^\top c(x) + \nabla \Psi_C(y^C). \quad (9)$$

HDVs use $G_H(y, h) = \Lambda_H^\top c(x)$.

For the exact response analysis, use a convex quadratic management objective:

$$\Psi_C(y^C) = \frac{1}{2} (y^C)^\top H_C y^C + q_C^\top y^C, \quad H_C = H_C^\top \succeq 0. \quad (10)$$

The matrix H_C is the curvature of the coalition's management objective on route flows, and q_C contains fixed marginal operating costs or route-specific priorities. Positive semidefiniteness is the modeling restriction that rules out locally concave management incentives; strict curvature is imposed only on the feasible OD-preserving tangent when a unique response is needed. The earlier perceived-slope model is nested by

$$H_C = \Lambda_C^\top \text{diag}(\widehat{\theta}^C) \Lambda_C, \quad q_C = 0. \quad (11)$$

Thus $\widehat{\theta}_e^C$ is an edge-separable parameterization of objective curvature, not the complete company objective. In particular, it should not be read as a measured coefficient of the BPR road function.

Under identity-preserving coordination, collect member edge flows on edge e in $\mathbf{f}_{C,e} = (f_e^i)_{i \in C}$. We parameterize governance by

$$\Psi_C(y^C) = \frac{1}{2} \sum_{e \in E} \mathbf{f}_{C,e}^\top \Omega_{C,e} \mathbf{f}_{C,e} + q_C^\top y^C, \quad \Omega_{C,e} = \Omega_{C,e}^\top \succeq 0. \quad (12)$$

The diagonal entries of $\Omega_{C,e}$ describe company-specific curvature. The off-diagonal entries describe the degree to which the coalition internalizes congestion interactions between members. This matrix is therefore a governance object: it changes whose marginal routing consequences are included in a decision, while the physical cost c_e remains unchanged.

For the main coalition-composition experiment, let $\tau_i > 0$ be a company objective type and let $\bar{b}_e \geq 0$ be a fixed reference curvature. We use

$$\Omega_{C,e}(\gamma) = \bar{b}_e D_C^{1/2} \left[(1 - \gamma) I_C + \gamma \mathbf{1}_C \mathbf{1}_C^\top \right] D_C^{1/2}, \quad D_C = \text{diag}(\tau_i)_{i \in C}, \quad \gamma \in [0, 1]. \quad (13)$$

The scalar γ is a governance parameter, not an information-quality estimate or an empirically calibrated technology parameter. At $\gamma = 0$, companies retain only their own diagonal management curvature. At $\gamma > 0$, the coalition internalizes cross-member interactions, with the same member types τ_i and reference slopes $\bar{b}_e$. The path is chosen so that increasing coordination adds off-diagonal terms without changing the sign of the curvature matrix; this makes comparative statics interpretable as a change in governance rather than a loss of well-posedness.

Proposition 2 (Convex governance path and inert partition labels). *The matrix $\Omega_{C,e}(\gamma)$ is positive semidefinite for every $\gamma \in [0, 1]$. Under identity-preserving constraints, if $\gamma = 0$ and $(\tau_i, q_i, d^i, \Lambda_i)$ are fixed, changing the coalition partition does not change the VI operator or its equilibrium set.*

Proof. For any vector v ,

$$v^\top \Omega_{C,e}(\gamma) v = \bar{b}_e(1 - \gamma) \|D_C^{1/2} v\|^2 + \bar{b}_e \gamma (\mathbf{1}_C^\top D_C^{1/2} v)^2 \geq 0.$$

At $\gamma = 0$, the management penalty is the sum of company-specific quadratic terms regardless of the coalition block in which each company is placed. Identity-preserving feasible sets are the same product of member-level OD polytopes for every partition, and the common physical term depends only on total edge flow. Therefore both the feasible set and stacked operator are unchanged. $\square$

3.4 Mixed Nash–Wardrop equilibrium

Let h be HDV route flow and define

$$\mathcal{Z}_\Pi = \{((y^C)_{C \in \Pi}, h) : y^C \in \mathcal{Y}_C, h \geq 0, \Gamma_H h = d^H\}. \quad (14)$$

Definition 1 (Heterogeneous-coalition mixed Nash–Wardrop equilibrium). *A feasible profile $z^* = ((y^{C*})_{C \in \Pi}, h^*)$ is an equilibrium for $(\Pi, \mathcal{G})$ if every coalition minimizes (8) over its feasible set given other populations, while HDVs use only minimum experienced-cost routes within each OD pair. Equivalently, if $\mathcal{F}_\Pi$ stacks G_C for all coalitions and G_H , then*

$$\mathcal{F}_\Pi(z^*)^\top (z - z^*) \geq 0 \quad \text{for all } z \in \mathcal{Z}_\Pi. \quad (15)$$

This is a finite-player Nash game coupled to a nonatomic Wardrop population. It is not assumed to possess a common potential: each coalition internalizes the marginal consequences specified by its own objective, whereas HDVs take the experienced road cost as given. The coalition block is a best-response condition because the objective is convex in its own flow; the HDV block is Wardrop’s equal-cost condition. The same physical cost c_e couples all populations, while H_C , q_C , Γ_C , and d^C encode organizational heterogeneity. The variational inequality is consequently a compact equilibrium representation, not an assertion that all agents maximize one social objective.

3.5 System benchmark and scope

Network congestion is measured by total system travel time

$$J(x) = \sum_{e \in E} x_e c_e(x_e). \quad (16)$$

Let J^{UE} be the all-nonatomic Wardrop benchmark and J^{SO} the system-optimal value for the same aggregate OD demand. We report normalized recovery

$$\rho = \frac{J^{UE} - J^{eq}}{J^{UE} - J^{SO}}. \quad (17)$$

The statistic measures the fraction of the recoverable congestion gap closed relative to the two benchmarks; it is not a complete platform-welfare measure and does not include operating profit, passenger surplus, reliability, or distributional effects.

For descriptive comparison only, company HHI is

$$\text{HHI}_{\text{firm}} = \sum_{i \in N} \left(\frac{m_i}{\sum_j m_j} \right)^2, \quad (18)$$

and coalition HHI is defined analogously using m_C . These indices summarize mass concentration but omit objective curvature, OD portfolios, route access, cross-member governance, and HDV adjustment.

Standing assumptions. The accepted demand batch, route sets, and physical costs are fixed within each counterfactual. Physical costs are continuous and nondecreasing. Each coalition management objective is continuously differentiable and convex in its own route flow. The main continuation theorem further assumes affine physical costs, a fixed active support, full-row-rank OD constraints, positive curvature on every OD-feasible coalition tangent, and a nonredundant HDV tangent after zero-slope directions are removed. These additional conditions are exactly the conditions needed for the projected linear response; they are not imposed on the BPR numerical game. Pricing, demand induction, passenger choice, capacity changes, and inter-batch repositioning remain outside scope.

4 Coalition Response and HDV Screening

The equilibrium definition specifies how coalitions choose routes, but it does not yet show which parts of their heterogeneous objectives reach aggregate congestion. The derivation follows the causal structure of the model. Existence and monotonicity establish that the continuation game is well posed. The coalition KKT system is then restricted to directions that preserve member OD obligations; this produces a projected response matrix rather than an unconstrained inverse of H_C . These response blocks are aggregated in edge space and screened by the HDV directions that can be reproduced through Wardrop reallocation. The resulting activated coalition signature is sufficient for a fixed-support affine equilibrium, while the BPR experiments remain numerical continuation results outside that closed form.

4.1 Existence and aggregate-flow uniqueness

Proposition 3 (Existence). *Under the standing assumptions in Section 3, every fixed coalition structure has at least one heterogeneous-coalition mixed Nash–Wardrop equilibrium.*

Proof. The feasible set $\mathcal{Z}_\Pi$ in (14) is a nonempty compact convex polytope. The physical costs are continuous, and the gradient of each convex management objective is continuous. Therefore the stacked operator $\mathcal{F}_\Pi$ is continuous on $\mathcal{Z}_\Pi$. The finite-dimensional Hartman–Stampacchia theorem gives a solution of the VI in (15). Convexity makes each coalition block equivalent to its best-response optimality condition, while the HDV block is Wardrop’s condition. $\square$

Proposition 4 (Monotonicity and unique congestible-edge flow). *Suppose physical costs are continuous and nondecreasing and the management objectives have the quadratic form (10). The stacked routing operator is monotone. If z^1, z^2 are equilibria, then*

$$x_e^1 = x_e^2$$

on every edge whose physical cost is strictly increasing between the two equilibrium flows. In particular, affine edges with $b_e > 0$ and nonconstant BPR edges have unique aggregate flow. If H_C is positive definite on $\ker \Gamma_C$ for each coalition, active coalition route flows are unique after duplicate-route coordinates are removed.

Proof. Let $\Delta y^C = y^{C,1} - y^{C,2}$, $\Delta h = h^1 - h^2$, and $\Delta x = \Lambda_H \Delta h + \sum_C \Lambda_C \Delta y^C$. Using (9) and (10),

$$\begin{aligned} & [\mathcal{F}_\Pi(z^1) - \mathcal{F}_\Pi(z^2)]^\top (z^1 - z^2) \\ &= \Delta x^\top [c(x^1) - c(x^2)] + \sum_{C \in \Pi} (\Delta y^C)^\top H_C \Delta y^C \geq 0. \end{aligned}$$

Each edge term is nonnegative because c_e is nondecreasing, and each quadratic term is nonnegative. Adding the two equilibrium variational inequalities forces this expression to be zero. On an edge where c_e is strictly increasing, the edge contribution can vanish only when $\Delta x_e = 0$. If H_C is positive definite on $\ker \Gamma_C$, feasibility implies $\Delta y^C \in \ker \Gamma_C$, and its zero quadratic form implies $\Delta y^C = 0$ in nonredundant coordinates. $\square$

Continuity and compact feasibility deliver existence; monotonicity, together with strict increase on congestible edges and tangent curvature where invoked, delivers the stated aggregate-flow and active-response uniqueness. These well-posedness statements are separate from the affine fixed-support representation below and from the BPR diagnostics in Section 6.

4.2 OD-restricted response of one coalition

Fix an active route support for coalition C . Restrict Λ_C and Γ_C to positive coalition routes and assume Γ_C has full row rank. A fixed support means that the currently positive routes are treated as the local coordinates; support inequalities are checked after solving. Let Z_C have orthonormal columns spanning $\ker \Gamma_C$. The relevant curvature condition is

$$v^\top H_C v > 0 \quad \text{for every } v \in \ker \Gamma_C \setminus \{0\}. \quad (19)$$

Lemma 1 (Coalition bordered nonsingularity). *The matrix*

$$\mathcal{B}_C = \begin{bmatrix} H_C & -\Gamma_C^\top \\ \Gamma_C & 0 \end{bmatrix} \quad (20)$$

is nonsingular if and only if $\ker H_C \cap \ker \Gamma_C = \{0\}$. For $H_C \succeq 0$, this is equivalent to (19).

Proof. If $\mathcal{B}_C(v, \mu)^\top = 0$, then $H_C v - \Gamma_C^\top \mu = 0$ and $\Gamma_C v = 0$. Premultiplying the first equation by $v^\top$ gives $v^\top H_C v = 0$. Positive semidefiniteness implies $v \in \ker H_C$, and the intersection condition gives $v = 0$. Full row rank then gives $\mu = 0$. Conversely, any nonzero $v \in \ker H_C \cap \ker \Gamma_C$ produces a null vector $(v, 0)$. $\square$

Define the projected inverse

$$P_C := Z_C (Z_C^\top H_C Z_C)^{-1} Z_C^\top. \quad (21)$$

Choose a feasible particular flow y_C^0 with $\Gamma_C y_C^0 = d^C$ and define

$$g_C := (I - P_C H_C) y_C^0 - P_C q_C. \quad (22)$$

This vector is independent of the particular flow because $P_C H_C Z_C = Z_C$.

Proposition 5 (OD-restricted coalition response). *On the fixed support, coalition C 's KKT equations imply*

$$y^C = g_C - P_C \Lambda_C^\top (a + Bx). \quad (23)$$

Consequently, the aggregate coalition response is

$$\sum_{C \in \Pi} \Lambda_C y^C = s_\Pi - R_\Pi (a + Bx), \quad (24)$$

where

$$R_\Pi := \sum_{C \in \Pi} \Lambda_C P_C \Lambda_C^\top, \quad s_\Pi := \sum_{C \in \Pi} \Lambda_C g_C. \quad (25)$$

The matrix R_Π is symmetric positive semidefinite.

Proof. On active routes, the coalition KKT system is

$$H_C y^C - \Gamma_C^\top \lambda_C = -\Lambda_C^\top (a + Bx) - q_C, \quad \Gamma_C y^C = d^C.$$

Write $y^C = y_C^0 + Z_C u$ and premultiply the first equation by $Z_C^\top$. Solving the resulting positive-definite system gives (23). Summing edge images gives (24). Since each P_C is symmetric positive semidefinite, the sum in (25) has the stated properties. $\square$

This result identifies exactly where objective heterogeneity enters. Differences in H_C matter only on the OD-preserving tangent, because directions normal to that tangent are ruled out by the demand equalities. Differences in q_C enter the intercept and therefore shift the baseline response. A change in a coalition label without a change in these objects or in its feasible constraints cannot change the response. This is why coalition size or HHI alone cannot serve as a sufficient state for the continuation equilibrium.

4.3 HDV screening

Let Z_H span $\ker \Gamma_H$ on the active HDV routes and define the HDV feasible edge tangent

$$L_H := \Lambda_H Z_H. \quad (26)$$

After removing directions that change flow only on constant-cost edges, assume

$$L_H^\top B L_H \succ 0. \quad (27)$$

Define

$$\mathcal{A}_H := I - L_H (L_H^\top B L_H)^{-1} L_H^\top B. \quad (28)$$

Lemma 2 (HDV screening projection). *The operator in (28) satisfies*

$$\begin{aligned} \mathcal{A}_H^2 &= \mathcal{A}_H, & \mathcal{A}_H L_H &= 0, & \ker \mathcal{A}_H &= \text{range}(L_H), \\ \text{range}(\mathcal{A}_H) &= \ker(L_H^\top B), & \mathcal{A}_H^\top B &= B \mathcal{A}_H. \end{aligned}$$

Proof. Let $K_H = L_H^\top B L_H$. Direct multiplication gives $\mathcal{A}_H L_H = 0$ and

$$\mathcal{A}_H^2 = I - 2L_H K_H^{-1} L_H^\top B + L_H K_H^{-1} K_H K_H^{-1} L_H^\top B = \mathcal{A}_H.$$

Thus $\text{range}(L_H) \subseteq \ker \mathcal{A}_H$. If $\mathcal{A}_H v = 0$, then $v = L_H K_H^{-1} L_H^\top B v$, so $v \in \text{range}(L_H)$. Moreover, $L_H^\top B \mathcal{A}_H = 0$, while $\mathcal{A}_H v = v$ whenever $L_H^\top B v = 0$; hence $\text{range}(\mathcal{A}_H) = \ker(L_H^\top B)$. Finally, symmetry of B and K_H gives $\mathcal{A}_H^\top B = B \mathcal{A}_H$. $\square$

4.4 Activated coalition-response theorem

Choose h^0 satisfying $\Gamma_H h^0 = d^H$ and write $\ell_H^0 = \Lambda_H h^0$. First remove zero-slope edge directions, so $B \succ 0$ on the retained edge space. Define the B -orthogonal representative space

$$\mathcal{Q}_H := \ker(L_H^\top B) = \text{range}(\mathcal{A}_H). \quad (29)$$

Let U_H have columns spanning $\mathcal{Q}_H$ and normalize them so that $U_H^\top B U_H = I$. Then $\mathcal{A}_H = U_H U_H^\top B$. This construction supplies explicit coordinates for the quotient by HDV-reallocation directions.

Theorem 1 (Activated coalition response and equilibrium equivalence). *Under the affine physical-cost specification, fixed coalition and HDV supports, and the conditions above, aggregate edge flow satisfies*

$$[I + \mathcal{A}_H R_\Pi B]x = \mathcal{A}_H(s_\Pi + \ell_H^0 - R_\Pi a) - L_H(L_H^\top B L_H)^{-1} L_H^\top a. \quad (30)$$

The coefficient is nonsingular on the retained positive-slope edge space. Hence all coalition objectives and the coalition partition enter the conditional aggregate equilibrium only through

$$\boxed{\mathfrak{S}_\Pi^{\text{act}} = (\mathcal{A}_H R_\Pi, \mathcal{A}_H s_\Pi)}. \quad (31)$$

Two governance arrangements with the same signature induce the same aggregate congestible-edge flow and TSTT whenever the common solution satisfies both arrangements' support inequalities. The support qualification is essential: when a route enters or leaves the active set, the local affine representation changes and a new signature must be computed.

Proof. Substitute (24) into $x = \Lambda_H h + \sum_C \Lambda_C y^C$ and write $h = h^0 + Z_H u$:

$$x = s_\Pi - R_\Pi(a + Bx) + \ell_H^0 + L_H u.$$

The HDV equal-cost equations, premultiplied by $Z_H^\top$, give $L_H^\top(a + Bx) = 0$. Eliminating u yields (30). Let $M_\Pi = I + \mathcal{A}_H R_\Pi B$ and suppose $M_\Pi v = 0$. Premultiplication by $I - \mathcal{A}_H$ gives $(I - \mathcal{A}_H)v = 0$, so $v \in \mathcal{Q}_H$. By the self-adjointness in Lemma 2,

$$v^\top B[I + \mathcal{A}_H R_\Pi B]v = v^\top Bv + (Bv)^\top R_\Pi(Bv) > 0,$$

because $B \succ 0$ and $R_\Pi \succeq 0$. Thus $v = 0$ and M_Π is nonsingular. The only organization-dependent terms are $\mathcal{A}_H R_\Pi$ and $\mathcal{A}_H s_\Pi$, proving sufficiency. $\square$

Let $x_H^a = -L_H(L_H^\top B L_H)^{-1} L_H^\top a$. Every edge flow satisfying the active HDV equal-cost equations has a unique representation $x = x_H^a + U_H z$. Define the reduced organization-dependent objects

$$\hat{R}_\Pi := U_H^\top B R_\Pi B U_H, \quad \hat{q}_\Pi := U_H^\top B \{s_\Pi - R_\Pi(a + Bx_H^a)\}. \quad (32)$$

Proposition 6 (Minimal reduced activated signature). *Fix the affine physical costs, coalition and HDV supports, and hence a common pair $(\mathcal{Q}_H, U_H)$. Let a common additive forcing $r \in \mathbb{R}^{\dim \mathcal{Q}_H}$ vary over a nonempty open set for which the support inequalities remain valid. Two governance arrangements generate the same reduced response map $z(r)$ on that set if and only if they have the same pair $(\hat{R}_\Pi, \hat{q}_\Pi)$.*

Proof. Substitution of $x = x_H^a + U_H z$ into the screened consistency equation and premultiplication by $U_H^\top B$ gives

$$z_\Pi(r) = (I + \hat{R}_\Pi)^{-1}(\hat{q}_\Pi + \hat{\ell}_H^0 + r), \quad \hat{\ell}_H^0 := U_H^\top B \ell_H^0,$$

where $\hat{\ell}_H^0$ is common across the arrangements. Equality on an open set implies equality of derivatives with respect to r , hence equality of $(I + \hat{R}_\Pi)^{-1}$ and then $\hat{R}_\Pi$. Equality at one forcing then gives equality of $\hat{q}_\Pi$. The converse follows directly from the displayed map. $\square$

The necessity statement concerns the reduced conditional response map on a common support, not one observed equilibrium and not the primitive matrices outside $\mathcal{Q}_H$. Different primitive objectives may produce the same reduced curvature and intercept, while a raw coalition displacement may be fully absorbed by HDVs. The reduced signature is therefore a minimal mechanism-level object in the precise coordinate system in which the response is observed.

4.5 Comparative statics of objective heterogeneity

Let η parameterize an objective, coalition composition, coordination rule, or feasible partition while the affine support and HDV tangent remain fixed. Define $K_\Pi = I + \mathcal{A}_H R_\Pi B$.

Proposition 7 (Objective-to-flow derivative). *If $R_\Pi(\eta)$ and $s_\Pi(\eta)$ are differentiable, then*

$$\frac{\partial x}{\partial \eta} = K_\Pi^{-1} \mathcal{A}_H \left[\frac{\partial s_\Pi}{\partial \eta} - \frac{\partial R_\Pi}{\partial \eta} (a + Bx) \right]. \quad (33)$$

For a fixed coalition tangent and curvature perturbation $\dot{H}_C$,

$$\dot{P}_C = -P_C \dot{H}_C P_C. \quad (34)$$

Under affine costs,

$$\frac{\partial J}{\partial \eta} = (a + 2Bx)^\top \frac{\partial x}{\partial \eta}. \quad (35)$$

Proof. Differentiate (30). The organization-dependent derivative on the right is $\mathcal{A}_H(\dot{s}_\Pi - \dot{R}_\Pi a)$, and the derivative of the left side contributes $K_\Pi \dot{x} + \mathcal{A}_H \dot{R}_\Pi Bx$. Rearranging gives (33). Differentiating (21) gives (34). Finally, $J(x) = a^\top x + x^\top Bx$ gives (35). $\square$

An objective or governance perturbation is locally irrelevant for aggregate flow when

$$\mathcal{A}_H \left[\dot{s}_\Pi - \dot{R}_\Pi (a + Bx) \right] = 0.$$

In particular, a displacement in $\text{range}(L_H)$ is completely absorbed by HDV reallocation. No universal sign for the TSTT derivative exists on a general network because the residual response can shift flow across multiple congested edges. Any sign reported in the numerical section is therefore conditional on the declared objective family, demand allocation, network, and support.

The derivative also provides a local decision rule for coalition design. Under identity-preserving coordination, a proposed merge of two existing blocks can be embedded in a differentiable path $\eta \in [0, 1]$ that activates their cross-member governance terms while preserving all member OD constraints. For such a path, define the merge-direction score

$$\mathcal{M}_\Pi(\eta) := (a + 2Bx)^\top K_\Pi^{-1} \mathcal{A}_H \left[\dot{s}_\Pi - \dot{R}_\Pi (a + Bx) \right]. \quad (36)$$

Proposition 8 (General-network merge regimes). *On a common affine support satisfying the conditions of Theorem 1, the score in (36) equals $dJ/d\eta$. Hence stronger coupling along the proposed merge path lowers, leaves unchanged, or raises TSTT according as $\mathcal{M}_\Pi(\eta)$ is negative, zero, or positive. If its sign is uniform on $[0, 1]$ and the support remains valid, the same ordering holds for the finite comparison between the two endpoints.*

Proof. Substituting (33) into (35) gives $dJ/d\eta = \mathcal{M}_\Pi(\eta)$. The local statements follow immediately. For the finite comparison, the fundamental theorem of calculus gives

$$J(1) - J(0) = \int_0^1 \mathcal{M}_\Pi(\eta) d\eta,$$

so a uniform sign determines the endpoint ordering. $\square$

This score does not impose a global lattice ordering on partitions. A merge can change the response direction as η varies, and a support change requires recomputing the score. It nevertheless replaces a purely ex post statement with a testable condition: centralization is favored along a candidate merge precisely when its OD-feasible response, after HDV screening, has negative system-cost projection.

4.6 Low-dimensional bottleneck reduction

The general construction reduces to a scalar expression on a two-route bottleneck. Let $c_1(x_1) = t_0 + t_1 x_1$ and $c_2 \equiv a$, and define $x^{UE} = (a - t_0)/t_1$. Let D contain coalitions that use both routes, and let H contain HDV bottleneck flow and coalition flow fixed at corners. For an interior coalition with perceived slope $\hat{\theta}_C = \beta_C t_1$, its bottleneck flow b_C satisfies

$$b_C = \frac{x^{UE} - x_1}{\beta_C}.$$

Summing gives

$$x_1 = \frac{H + \Sigma x^{UE}}{1 + \Sigma}, \quad \Sigma = \sum_{C \in D} \frac{1}{\beta_C}. \quad (37)$$

Within a fixed responsive set,

$$\frac{\partial x_1}{\partial \Sigma} = \frac{x^{UE} - H}{(1 + \Sigma)^2}, \quad \frac{\partial x_1}{\partial H} = \frac{1}{1 + \Sigma}.$$

Let total two-route demand be N . Up to the constant Na , system travel time on this support is

$$J(\Sigma) = Na + t_1 \{x_1(\Sigma)^2 - x^{UE} x_1(\Sigma)\}. \quad (38)$$

Theorem 2 (Canonical coalition-design regimes). *Consider the two-route network above and a fixed coalition objective family. If HDVs use both routes in the interior, their Wardrop equality pins $x_1 = x^{UE}$, so coalition organization is locally neutral for aggregate flow and TSTT. Otherwise suppose the responsive support is described by (37), with $0 \leq H < x^{UE}$ and $\Sigma \geq 0$. Then*

$$\frac{dJ}{d\Sigma} = \frac{t_1(x^{UE} - H)[2H + (\Sigma - 1)x^{UE}]}{(1 + \Sigma)^3}. \quad (39)$$

Consequently:

1. if $H \geq x^{UE}/2$, J is nondecreasing in Σ , and every organizational change that lowers effective response improves TSTT;
2. if $H < x^{UE}/2$, J decreases for $\Sigma < \Sigma^*$ and increases for $\Sigma > \Sigma^*$, where

$$\Sigma^* = 1 - \frac{2H}{x^{UE}}; \quad (40)$$

among attainable organizations on the same support, the preferred one moves effective response toward Σ^* .

Thus a grand coalition is preferred when its effective response lies closest to the relevant boundary or interior target, whereas a partial coalition can strictly dominate both grand and singleton organization when it is closer to Σ^ than either endpoint.*

Proof. Substitute (37) into (38) and use

$$\frac{dJ}{d\Sigma} = t_1(2x_1 - x^{UE})\frac{dx_1}{d\Sigma}.$$

Because

$$2x_1 - x^{UE} = \frac{2H + (\Sigma - 1)x^{UE}}{1 + \Sigma},$$

equation (39) follows. Under $0 \leq H < x^{UE}$, every factor outside the bracket is positive. The bracket has no nonnegative root when $H \geq x^{UE}/2$ and has the unique root $\Sigma^* \in (0, 1]$ otherwise, which proves the stated regimes. If HDVs are interior on both routes, Wardrop equality gives $t_0 + t_1x_1 = a$ and hence $x_1 = x^{UE}$ independently of coalition response. $\square$

The theorem answers the organizational question without claiming that every merger moves Σ in the same direction. In the screened regime, organization has no first-order congestion effect. In the monotone regime, a merge is desirable when it lowers effective response. In the interior regime, neither maximal centralization nor maximal fragmentation is generically optimal; the relevant target is Σ^* . The scalar pair (H, Σ) is sufficient only because this network has one feasible diversion margin. Once a coalition has several interacting routes or several OD obligations, R_Π is generally a matrix with cross-edge entries, and Proposition 8 supplies the corresponding network-level criterion.

5 Coalition Structure and Objective Composition

The model separates three organizational dimensions that are often compressed into one market-share statistic. The underlying company count n determines how many heterogeneous management objectives exist. The coalition count $K = |\Pi|$ determines how many independent routing decision centers remain. The coalition-size and objective-type composition determine which cross-member interactions are internalized. HHI is a descriptive summary of mass concentration at either the company or coalition level; it does not determine these governance objects.

5.1 What a coalition changes

For identity-preserving coordination, coalition formation leaves every member's accepted OD obligation intact. Its direct equilibrium effect comes from the off-diagonal elements of $\Omega_{C,e}$ and from any governance weights applied to member objectives. A coalition therefore changes the curvature matrix H_C in the KKT response, and hence changes P_C , R_Π , and the activated response. The partition itself is inert when all off-diagonal terms are zero, as established in Proposition 2.

Operational pooling is a different intervention. It relaxes member-level conservation constraints and can enlarge the coalition's feasible tangent. Proposition 1 shows that this enlargement may disappear at the aggregate level when members have identical route sets and the objective depends only on total coalition flow. The empirical design keeps pooling separate so that the composition and degree results identify objective coordination rather than a demand-reassignment channel.

5.2 Objective composition at fixed coalition size

Consider four equal-mass companies with objective types (L_1, L_2, H_1, H_2) , where L and H denote low and high management curvature. The assortative partition

$$\Pi_{\text{assort}} = \{\{L_1, L_2\}, \{H_1, H_2\}\}$$

and the mixed partition

$$\Pi_{\text{mixed}} = \{\{L_1, H_1\}, \{L_2, H_2\}\}$$

have the same number of coalitions, the same coalition sizes, the same company masses, and the same company and coalition HHI. They differ only in how heterogeneous objectives are grouped. Under $\gamma > 0$, the two partitions generate different cross-member governance matrices and therefore different activated responses. This is the cleanest comparison for the paper's central question.

5.3 Coalition degree and company count

The degree of organization has two levels. Holding n fixed, decreasing K from n to one aggregates more companies into each routing decision center. Holding total AV mass fixed while changing n changes both the number of heterogeneous objective holders and the mass of each company. These comparisons answer different questions and are reported separately. A change in HHI across n is a consequence of equal-mass design, not an interpretation of the coalition mechanism.

The degree scan uses $n \in \{2, 4, 8\}$ and balanced partitions. At full coordination, the grand-minus-singleton effect can change sign across n and AV penetration. Thus neither K nor n supplies a universal congestion ordering. The network's route geometry and the HDV tangent determine how the coalition response is activated.

5.4 Coalition structure as a state-contingent design variable

The continuation model also defines a practical outer design problem. Let $x^*(\Pi, \mathcal{G}; \alpha, \theta)$ denote the certified equilibrium induced by partition Π , governance rule $\mathcal{G}$, AV penetration α , and objective-type vector θ . A network manager who can select from a declared menu of coordination arrangements faces the conditional problem

$$\Pi^*(\alpha, \theta, \mathcal{G}) \in \arg \min_{\Pi \in \mathfrak{P}(N)} J(x^*(\Pi, \mathcal{G}; \alpha, \theta)), \quad (41)$$

where $\mathfrak{P}(N)$ is the set of feasible partitions. This is a continuation-design problem, not an endogenous formation game: it evaluates the congestion consequences of each specified arrangement without asserting that firms would voluntarily choose the minimizer. Because numerical equilibria can be tied up to certification tolerance, the empirical implementation uses the set-valued rule

$$\mathfrak{P}_\varepsilon^*(s) := \{\Pi : J(\Pi; s) - J^*(s) \leq \varepsilon J^{UE}\}, \quad J^*(s) := \min_{\Pi \in \mathfrak{P}(N)} J(\Pi; s). \quad (42)$$

At $\alpha = 0.5$ and $\gamma \in \{0.5, 1\}$, the grand coalition is the unique best arrangement in the 15-partition Sioux Falls menu. At $\alpha = 0.9$, the best set contains symmetry-equivalent three-one partitions at $\gamma = 0.5$ and a two-one-one partition at $\gamma = 1$. The preferred organizational granularity therefore changes with the state.

Coordination also consumes managerial, informational, contracting, or computational resources. To separate this burden from congestion performance, define the normalized governance burden

$$C_{\text{gov}}(\Pi, \gamma) := \gamma^2 \frac{\sum_{C \in \Pi} \binom{|C|}{2}}{\binom{n}{2}} \quad (43)$$

and the penalized continuation loss

$$\mathcal{L}_\lambda(\Pi; s) := \frac{J(\Pi; s) - J^{SO}}{J^{UE} - J^{SO}} + \lambda C_{\text{gov}}(\Pi, \gamma), \quad \lambda \geq 0. \quad (44)$$

The burden in (43) is a transparent design primitive, not an empirical estimate. Its pair count records the fraction of company pairs that require cross-member governance, while γ^2 makes stronger coupling more costly.

Proposition 9 (Governance-cost threshold). *Fix a state s . Consider arrangements A and B such that $J_A < J_B$ and $C_A > C_B$, where $C_A = C_{\text{gov}}(A, \gamma)$ and similarly for B . Their penalized losses are equal at*

$$\lambda_{A,B}^* = \frac{J_B - J_A}{(J^{UE} - J^{SO})(C_A - C_B)}. \quad (45)$$

Arrangement A is preferred for $\lambda < \lambda_{A,B}^$, and B is preferred for $\lambda > \lambda_{A,B}^*$. The optimal organization over a finite menu is the lower envelope of these affine losses and is therefore piecewise constant in λ .*

Proof. Subtract (44) for the two arrangements. The common system-optimum term cancels, leaving

$$\mathcal{L}_\lambda(A; s) - \mathcal{L}_\lambda(B; s) = \frac{J_A - J_B}{J^{UE} - J^{SO}} + \lambda(C_A - C_B).$$

This affine difference crosses zero once at (45); its slope is positive. Taking the pointwise minimum across a finite collection gives the lower-envelope result. $\square$

The proposition yields grand, partial, and fragmented governance regimes without treating enumeration itself as the scientific conclusion. Network equilibrium determines the intercept of each arrangement's loss, governance burden determines its slope, and the pairwise thresholds determine where the recommended organization changes.

5.5 Why HHI remains auxiliary

For company shares $s_i = m_i / \sum_j m_j$, company HHI is $\sum_i s_i^2$. For coalition shares $s_C = m_C / \sum_D m_D$, coalition HHI is $\sum_C s_C^2$. These scalars record only the distribution of mass. They do not record:

1. the objective curvature H_C or fixed marginal terms q_C ;
2. the OD constraints Γ_C and the routes available to each member;
3. the cross-member governance terms in $\Omega_{C,e}$;
4. the HDV adjustment space $\text{range}(L_H)$;
5. the active support on which the response is valid.

The fixed-HHI spatial experiment in Section 6 supplies a direct network witness: equal company shares and equal HHI can still produce materially different congestion when spatial OD portfolios change. HHI is therefore useful as a control descriptor, but it is not the paper's equilibrium state variable.

5.6 Relation to endogenous coalition formation

The main paper studies the equilibrium consequences of specified coalition structures. Endogenous formation is a separate partition-function game because a coalition’s cost depends on the full partition and because its cost must be allocated among member companies. A stability claim would require an equilibrium-selection rule, a transfer or payoff-allocation rule, and a deviation protocol. Without these primitives, comparing the continuation equilibria of two partitions is not evidence that either partition is stable or that firms would choose it.

6 Experiments: From Coalition Mechanism to Design Map

The computational analysis is organized around a single question: when the physical network and aggregate demand are held fixed, when should routing authority be centralized and when should some independent decision centers be retained? We proceed from mechanism-isolating networks to a certified coalition-design policy. The bottleneck verifies the analytic regime threshold; Sioux Falls supplies a realistic benchmark for objective composition and governance-cost regimes; and independent Braess and grid topologies test whether the state transition and local merge policy survive a change in network geometry. Complete $n = 4$ partition scans are used to measure policy regret, not as the scientific conclusion. Coalition partitions are counterfactual inputs throughout, so the experiments identify conditional equilibrium consequences rather than endogenous formation incentives.

6.1 Experimental design and certification

Every counterfactual keeps aggregate OD demand, physical link costs, and admissible routes fixed. In the identity-preserving treatment, each company retains its own accepted OD obligation, so a partition changes the aggregation of management objectives and the number of decision centers without changing the demand that must be served. The company-count scan holds total AV mass fixed while changing the number of underlying companies n ; company HHI is reported descriptively and is not treated as a causal estimand. We report total system travel time (TSTT),

$$J(x) = \sum_{e \in E} x_e c_e(x_e),$$

and the normalized recovery ρ in (17). Because the user-equilibrium and system-optimum benchmarks are fixed within a comparison, maximizing recovery is equivalent to minimizing TSTT. Recovery is a normalized congestion-gap measure, not a percentage reduction in TSTT.

The finite-route variational inequality is solved with an adaptive Korpelevich extragradient method, simplex projection for each company–OD block, and player-specific shortest-path column generation. A profile is retained only when the normalized VI gap is at most 10^{-6} and the full-network omitted-route slack is at most 10^{-6} of the route-cost scale. The composition scan has 40 certified rows with maximum VI gap 9.9998479×10^{-7} and maximum relative omitted-route slack 9.9227901×10^{-7} ; the degree scan has 54 certified rows with corresponding maxima 9.9749371×10^{-7} and 9.9064955×10^{-7} . The larger sensitivity, complete-partition, and pooling packages are certified under the same criteria.

6.2 Mechanism tests: from scalar backfilling to screened responses

Bottleneck. Consider one affine congestible link $c_1(x_1) = t_0 + t_1 x_1$ and a constant alternative $c_2 \equiv a$. On a fixed support, an interior coalition with perceived slope $\hat{\theta}_C = \beta_C t_1$ sends $b_C =$

$(x^{UE} - x_1)/\beta_C$ on the bottleneck. Summing across responsive coalitions gives

$$x_1 = \frac{H + \Sigma x^{UE}}{1 + \Sigma}, \quad \Sigma = \sum_{C \in D} \frac{1}{\beta_C}, \quad (46)$$

where H collects HDV flow and coalition flow fixed at a corner. The equation makes the mechanism transparent: governance changes the aggregate coalition response through Σ , while HDV backfilling determines how much of that response reaches aggregate flow. For the canonical demand split, HDVs pin $x_1 = x^{UE}$ below the support boundary $\alpha_c = 1 - x^{UE}/N$; above the boundary, the alternative HDV route leaves support and the coalition displacement becomes active. This is an exact scalar reduction, not a claim that general networks have a one-dimensional response.

Braess. The Braess network uses demand $D = 4000$, congested-edge cost $x_e/100$, outer-link costs 45, a zero-cost shortcut, and three active fleet routes. The unrestricted route-space curvature is singular, but its restriction to OD-preserving directions is positive definite, so the projected coalition response is well defined. The response matrix has off-diagonal entries because changing one edge flow shifts mass across other edges used by the same OD routes. When all three HDV routes are active, the HDV tangent spans the coalition response and the activated operator is zero. Removing the shortcut from the HDV support removes one adjustment direction; the same coalition-side response then retains rank one. This pair of panels isolates the two transformations used in the theory: OD projection on the coalition side and HDV screening on the network side.

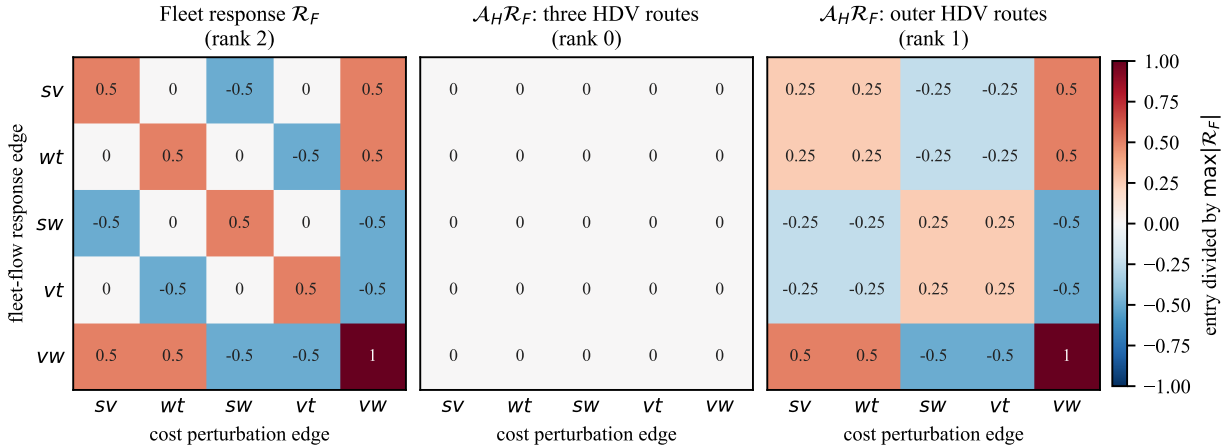


Figure 1: Raw coalition response and HDV screening on the affine Braess network. Entries are normalized by the largest raw response entry. The coalition-side operator is held fixed across panels; changing only the HDV support changes the rank of the activated response.

6.3 Sioux Falls: benchmarks and the conditional design criterion

Sioux Falls has 76 directed links and 528 positive-demand OD pairs. The BPR benchmarks are $J^{UE} = 7,480,226$ and $J^{SO} = 7,194,256$, leaving a recoverable gap of 285,970. Let $\mathfrak{P}(N)$ denote the set of partitions in the declared design menu and let $x^*(\Pi; \alpha, \gamma, \theta)$ be the certified continuation equilibrium under partition Π , AV penetration α , coordination intensity γ , and objective-type vector θ . The conditional design problem evaluated below is

$$\Pi^*(\alpha, \gamma, \theta) \in \arg \min_{\Pi \in \mathfrak{P}(N)} J(x^*(\Pi; \alpha, \gamma, \theta)). \quad (47)$$

Equation (47) is an ex post design comparison over specified arrangements. It does not assign firms a payoff, permit deviations, or claim that the selected partition is stable.

6.4 Objective composition at fixed concentration

The first Sioux Falls experiment holds constant the quantities that a concentration statistic can see. Four equal-mass companies have objective types

$$(\tau_1, \tau_2, \tau_3, \tau_4) = (0.5, 0.5, 1.5, 1.5),$$

proportional but separately conserved OD obligations, company HHI 0.25, two equal-mass coalitions, coalition HHI 0.5, and identical route access. We compare assortative pairs $\{\{1, 2\}, \{3, 4\}\}$ with mixed pairs $\{\{1, 3\}, \{2, 4\}\}$. Only the composition of objective types within the coalitions changes.

Table 1: Certified assortative-minus-mixed composition contrasts. Positive recovery differences favor assortative grouping; positive $\Delta J/J^{UE}$ means higher assortative TSTT.

γ	$\alpha = 0.5$		$\alpha = 0.9$	
	Δ recovery (pp)	$\Delta J/J^{UE}$	Δ recovery (pp)	$\Delta J/J^{UE}$
0	0.000	0.0000%	0.000	0.0000%
0.25	-0.294	0.0112%	-1.285	0.0491%
0.50	-1.168	0.0446%	-1.161	0.0444%
0.75	-0.244	0.0093%	-0.228	0.0087%
1.00	1.876	-0.0717%	1.031	-0.0394%

At $\gamma = 0$, the two partitions are invariant up to solver tolerance, as predicted by Proposition 2. Once cross-member terms are active, the ordering changes with governance: mixed pairs close more of the recoverable gap at $\gamma = 0.25$ – 0.75 , whereas assortative pairs are better at $\gamma = 1$. The largest baseline TSTT contrast is 0.0717% of J^{UE} , so the effect is modest in absolute network time but decisive for the ordering of otherwise matched arrangements. This is a controlled identification result: an average objective type, coalition size, or HHI cannot recover the relevant response.

6.5 State-contingent regimes in Sioux Falls

We evaluate all 15 set partitions of the same four companies so that the performance of any proposed selection rule can be measured against a known benchmark. With $\varepsilon = 10^{-8}$ in (42), Table 2 reports the set of certified optima rather than forcing a numerical tie to be unique.

Table 2: Set-valued optimum in the Sioux Falls $n = 4$ audit scan. Company labels are one based. All comparisons use the same objective types, OD portfolios, route access, and identity-preserving governance rule.

α	γ	ε -optimal partition set	K	coalition HHI	gain vs. singleton (pp)
0.5	0.5	$\{\{1, 2, 3, 4\}\}$	1	1.000	18.096
0.5	1.0	$\{\{1, 2, 3, 4\}\}$	1	1.000	18.810
0.9	0.5	$\{\{1, 2, 3\} \{4\}, \{1, 2, 4\} \{3\}\}$	2	0.625	8.745
0.9	1.0	$\{\{1, 2\} \{3\} \{4\}\}$	3	0.375	8.540

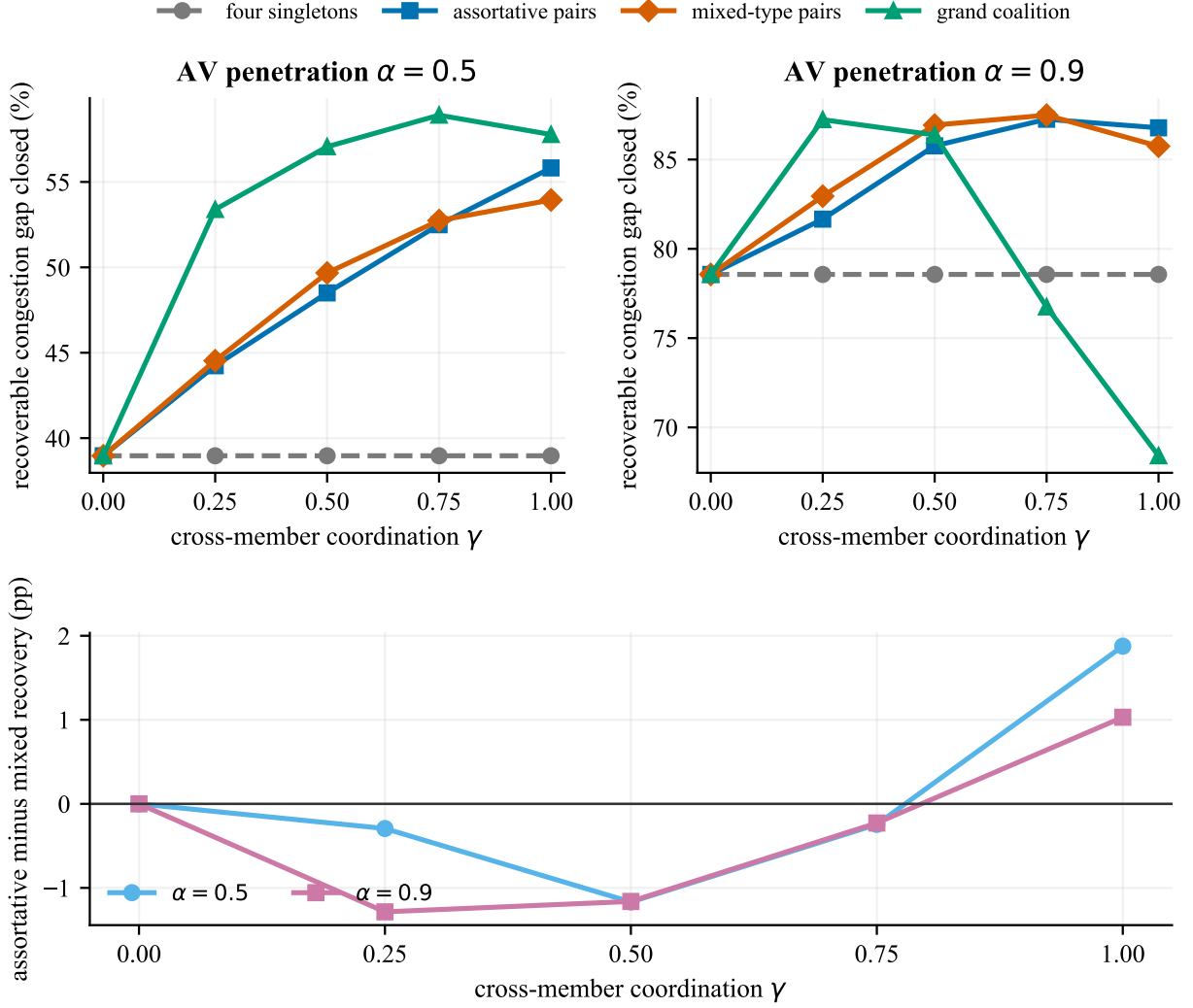


Figure 2: Coalition objective composition at fixed company masses, coalition sizes, OD obligations, and HHI. The top panels report recovery by partition; the bottom panel reports the assortative-minus-mixed recovery contrast.

The benchmark exhibits a clear state transition. At moderate AV penetration, the grand coalition is optimal for both tested governance intensities. At high penetration, partial coordination is optimal. The two three-one arrangements at $\alpha = 0.9, \gamma = 0.5$ differ in TSTT by only 5.58×10^{-7} , or $7.46 \times 10^{-12} J^{UE}$, because the two high-type companies are symmetric. At $\gamma = 1$, the optimum retains three decision centers. Thus stronger governance does not imply that all companies should share one dispatcher. This pattern is consistent with Theorem 2: coordination should move the network toward its effective response target, which may lie at an endpoint or in the interior.

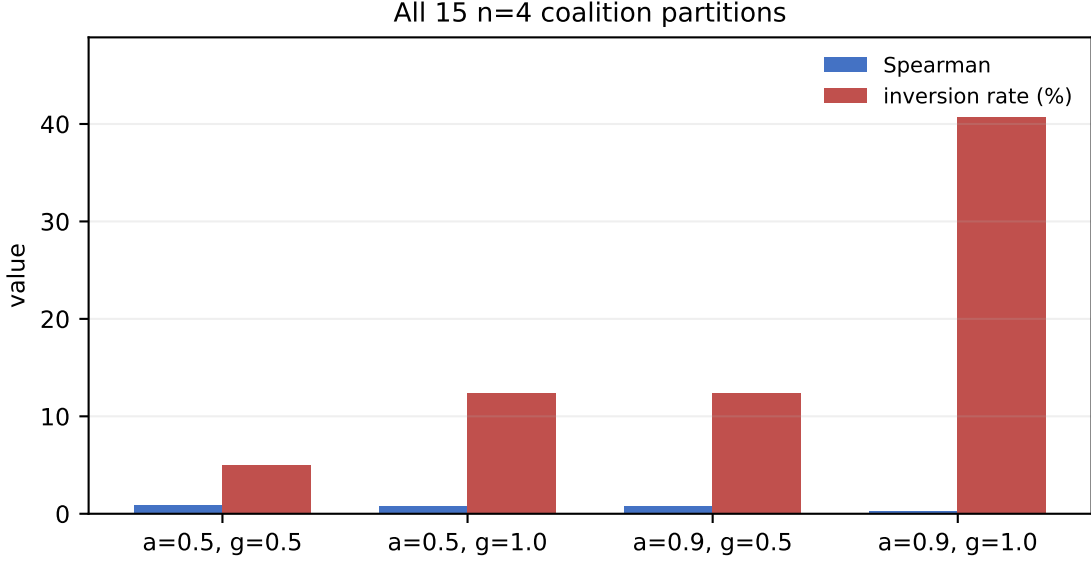


Figure 3: Complete $n = 4$ Sioux Falls audit scan. The exhaustive values establish the benchmark optimum used to evaluate local selection rules; they are not required by the rules themselves.

6.6 Local selection rules and policy regret

For larger company sets, enumerating the Bell-number partition space is not a practical design rule. We therefore test two agglomerative policies. The one-step policy starts from singleton companies, evaluates every feasible pairwise merge of the current blocks, chooses the best improving merge, and stops when none improves TSTT. The two-step policy chooses the first merge on the best improving path of length at most two. Both policies use only local coarsenings of the current partition. The complete $n = 4$ scans are retained solely to calculate their ex post recovery regret

$$\text{Reg}(\hat{\Pi}; s) := 100 \frac{J(\hat{\Pi}; s) - J^*(s)}{J^{UE} - J^{SO}} \quad \text{recovery percentage points.} \quad (48)$$

The one-step rule is exact in all Sioux Falls states and in 29 of 32 cross-network states, with mean regret 0.332 points, but it has an 8.339-point failure on Braess at $\alpha = 0.7, \gamma = 0.75$. The other failures are 2.141 points at $(0.9, 0.5)$ and 0.130 points at $(0.9, 1)$. Two-step lookahead repairs these local traps and reaches an ε -optimal arrangement in all 36 tested states. This is benchmark evidence, not a theorem of universal optimality. For four companies the two-step neighborhood covers much of the small partition lattice; for general n it remains a local rule and avoids requiring an ex ante enumeration of all Bell-number partitions.

Table 3: Policy regret against the complete $n = 4$ audit scans. Exact counts use the same relative TSTT tolerance as (42).

benchmark	policy	exact states	mean regret (pp)	max regret (pp)
Sioux Falls (4 states)	one-step merge	4/4	0.000	0.000
	two-step merge	4/4	0.000	0.000
	always grand	2/4	4.908	18.695
	always singleton	0/4	13.548	18.810
Braess and grid (32 states)	one-step merge	29/32	0.332	8.339
	two-step merge	32/32	0.000	0.000
	always grand	27/32	1.520	18.205
	always singleton	0/32	21.752	42.289

6.7 Governance-cost regimes

The TSTT-minimizing organization need not be preferred once coordination requires institutional resources. We apply (44) to the Sioux Falls audit scan. At $\alpha = 0.5, \gamma = 1$, the lower envelope passes through all four organizational degrees as λ increases.

Table 4: Governance-cost regimes at $\alpha = 0.5, \gamma = 1$. Endpoints are pairwise indifference thresholds; the displayed partition is optimal in the open interval.

λ interval	selected partition	description	K
$[0, 0.0293)$	$\{1, 2, 3, 4\}$	grand coalition	1
$[0.0293, 0.2678)$	$\{1, 2\} \{3, 4\}$	two pairs	2
$[0.2678, 0.7435)$	$\{1, 2\} \{3\} \{4\}$	one pair	3
$[0.7435, \infty)$	$\{1\} \{2\} \{3\} \{4\}$	singletons	4

The sequence is not imposed by a preference for intermediate structures. Each partition supplies an equilibrium-congestion intercept and a governance-cost slope; Proposition 9 determines their crossings. When coordination is inexpensive, the congestion gain justifies a grand coalition. As the burden rises, partial coalitions retain the most valuable internalization while avoiding some cross-company links. Fragmented control becomes optimal only after the congestion benefit no longer covers the governance burden.

6.8 Cross-network validation of the regime transition

We repeat the complete 15-partition audit on two independent BPR topologies: a five-edge Braess network and a directed 3×3 grid with three ODs. The design crosses four penetrations $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$, four governance intensities $\gamma \in \{0.25, 0.5, 0.75, 1\}$, and the common objective vector $(0.5, 0.5, 1.5, 1.5)$. All $480 = 2 \times 4 \times 4 \times 15$ profiles pass the VI and full-route oracle tests. The maximum VI gap is 9.971×10^{-7} and the maximum relative omitted-route slack is 3.79×10^{-16} .

Table 5: Cross-network organizational regimes over 16 states per topology.

network	grand-optimal states	partial-optimal states	transition at $\alpha = 0.9$
Braess	14	2	partial for $\gamma \in \{0.75, 1\}$
directed grid	13	3	partial for $\gamma \in \{0.5, 0.75, 1\}$

Both networks select the grand coalition in every tested state with $\alpha \leq 0.7$. At $\alpha = 0.9$, sufficiently strong governance produces a partial-coordination regime on both topologies, although the transition occurs at a lower γ on the grid. The qualitative state transition therefore survives a change in topology, OD count, and cost calibration, while its boundary remains network dependent.

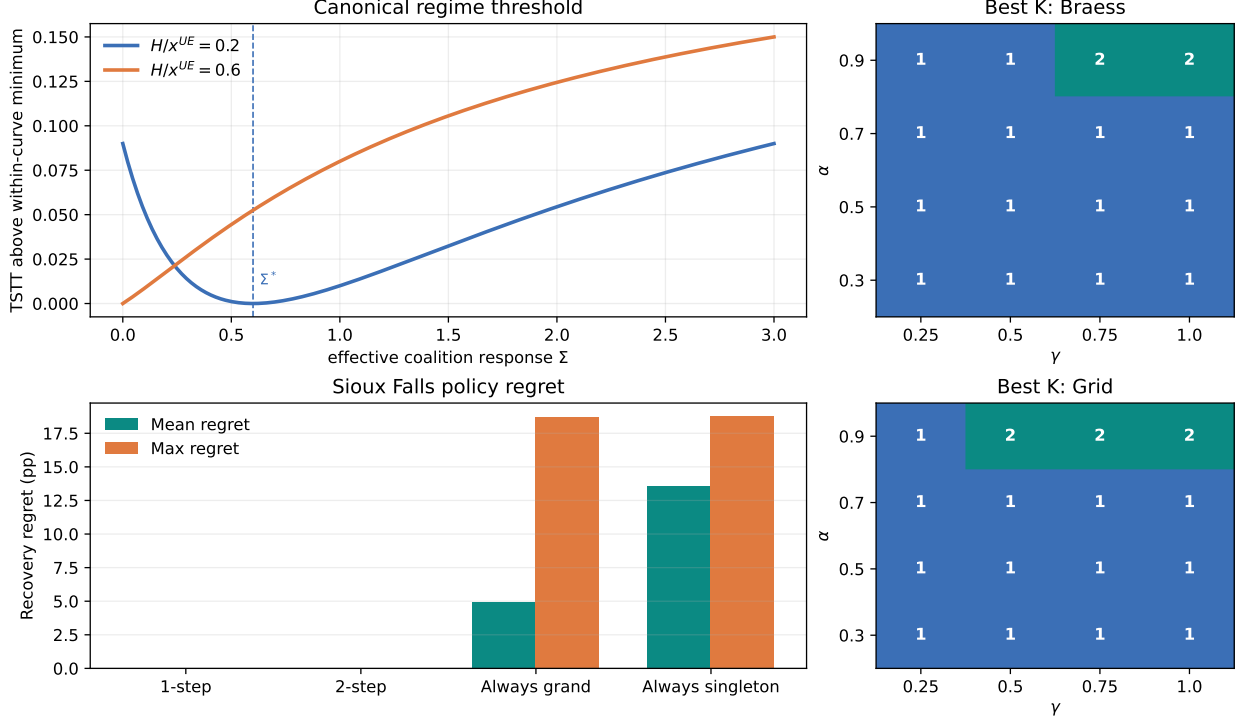


Figure 4: Regime theory and validation. The canonical bottleneck has an interior target Σ^* only when $H < x^{UE}/2$; the Braess and grid heatmaps report the best coalition count; the Sioux Falls bars report policy regret.

6.9 Company count and coalition degree

The degree scan varies the number of underlying companies $n \in \{2, 4, 8\}$ while holding total AV mass and aggregate demand fixed. Companies have equal mass within each n and alternate low and high objective types. For each n , singleton, balanced intermediate, and grand-coalition partitions are compared within the same objective-type distribution.

Table 6: Grand-minus-singleton coalition-degree effects at full cross-member coordination. Company HHI is descriptive; K and n are distinct organizational dimensions.

α	n	$\rho_{K=1} - \rho_{K=n}$ (pp)	$\Delta J/J^{UE}$	company HHI
0.5	2	1.956	-0.075%	0.500
0.5	4	18.811	-0.719%	0.250
0.5	8	29.489	-1.127%	0.125
0.9	2	-18.365	0.702%	0.500
0.9	4	-10.155	0.388%	0.250
0.9	8	11.927	-0.456%	0.125

At $\alpha = 0.5$, the grand coalition improves recovery relative to singleton governance, with the effect increasing from 1.956 to 29.489 points as n rises from 2 to 8. At $\alpha = 0.9$, the same intervention is harmful for $n = 2$ and $n = 4$ but beneficial for $n = 8$, ranging from -18.365 to 11.927 points. Intermediate coordination is also material: at $\alpha = 0.9$, moving from singletons to balanced pairs changes recovery by 7.179 points for $n = 4$ and 25.227 points for $n = 8$. The comparison makes clear why an ownership statistic cannot substitute for reporting both the underlying company count and the active coalition count.

Coalition degree at full cross-member coordination ($\gamma = 1$)

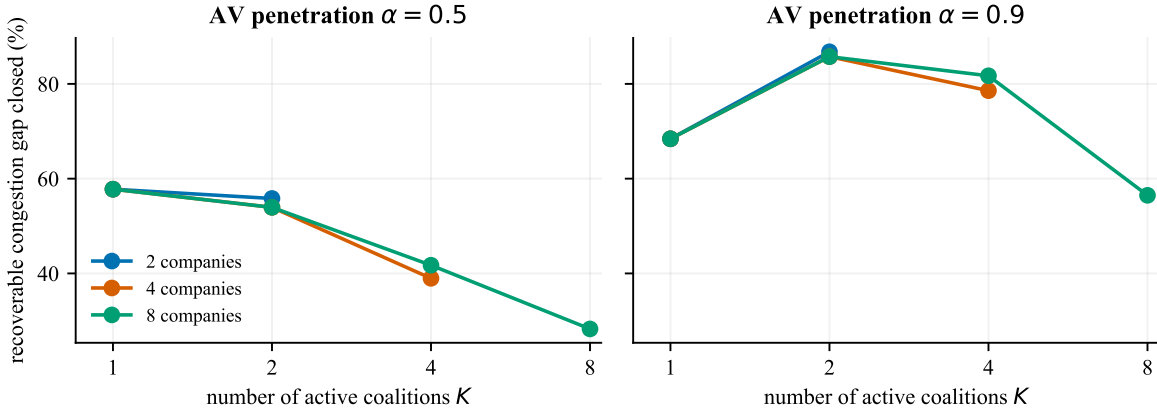


Figure 5: Coalition degree at full cross-member coordination. Lines connect balanced partitions across the available coalition counts K for each underlying company count n .

6.10 Sensitivity, audit diagnostics, and weak-effect boundaries

The controlled composition comparison is repeated over objective-type gaps $\tau_H - \tau_L \in \{0.5, 1.0, 1.5\}$, reference management-slope scales $\{0.5, 1, 2\}\bar{b}$, five values of γ , and $\alpha \in \{0.5, 0.9\}$. All 180 profiles are certified. The two partitions coincide at $\gamma = 0$ for every design. For $\gamma > 0$, the contrast is generally nonzero but is not sign invariant. The absolute contrast reaches 9.089 recovery points (wide gap, $\alpha = 0.9$, unit slope scale). Sign reversals over γ occur in multiple gap-scale combinations, including the baseline design. The robust statement is therefore behavioral: objective composition is active under coordination. No universal ranking of assortative and mixed grouping is supported.

Table 7: Sensitivity summary for the assortative-minus-mixed recovery contrast. Each row aggregates five γ values and is based on certified profiles.

α	type gap	slope scale	min contrast (pp)	max contrast (pp)	sign reversal
0.5	0.5	0.5	-0.496	0.210	yes
0.5	1.0	1.0	-1.168	1.877	yes
0.5	1.5	1.0	-3.553	0.000	no
0.9	0.5	1.0	-1.029	0.018	yes
0.9	1.0	1.0	-1.285	1.031	yes
0.9	1.5	1.0	-9.089	0.000	no

The complete $n = 4$ scan also audits whether a mass statistic could substitute for the response-based policy. All 15 partitions are evaluated at $\alpha \in \{0.5, 0.9\}$ and $\gamma \in \{0.5, 1\}$, yielding 60 certified profiles. At $\alpha = 0.5$, coalition-HHI/recovery Spearman correlations are 0.906 and 0.793 at $\gamma = 0.5$ and 1, with inversion rates 4.94% and 12.35%. At $\alpha = 0.9$, the correlation falls from 0.815 to 0.236 and the inversion rate rises from 12.35% to 40.74%. The high-penetration, full-coordination profiles span 0.684–0.871 in recovery and 0.715% of J^{UE} in TSTT. These are systematic within-benchmark diagnostics, not population estimates. They explain why HHI is inadequate for selecting among the candidate merges evaluated by the local policies.

Table 8: All $n = 4$ set-partition scan. Values are computed across the 15 partitions in each condition.

α	γ	Spearman	Kendall	inversion rate	recovery range
0.5	0.5	0.906	0.807	4.94%	0.390–0.571
0.5	1.0	0.793	0.671	12.35%	0.390–0.578
0.9	0.5	0.815	0.671	12.35%	0.786–0.873
0.9	1.0	0.236	0.165	40.74%	0.684–0.871

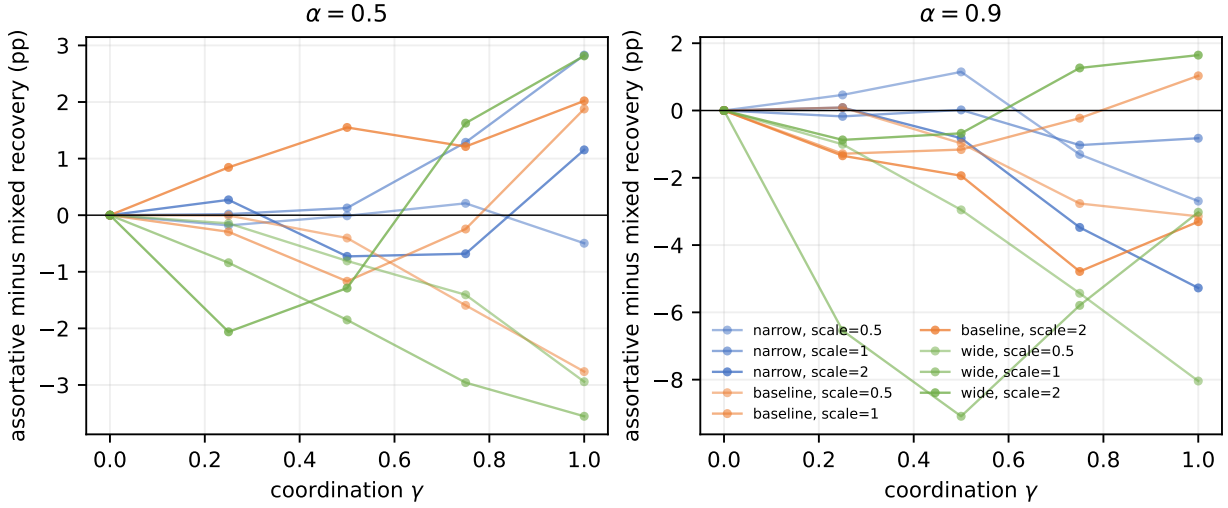


Figure 6: Objective-gap and governance-scale sensitivity. Lines report the certified assortative-minus-mixed recovery contrast over γ ; colors denote objective gap and line opacity denotes reference-slope scale.

6.11 Spatial exposure and HDV penetration

The fixed-HHI spatial experiment assigns four equal fleets the same total OD demand and company HHI 0.25 but changes their spatial OD portfolios. Recovery spans 30.75%–42.44% at $\alpha = 0.5$ and 10.15%–74.33% at $\alpha = 0.9$. The largest displacement from proportional allocation is 0.363% and 2.360% of J^{UE} , respectively. Spatial exposure is therefore a design input alongside coalition size: identical masses can have different activated responses because they control different OD corridors.

The penetration scan under common $\beta = 1$ reports all-pair reversal rates of 18.10%, 18.73%, 10.65%, 10.14%, and 6.75% at $\alpha = 0.1, 0.3, 0.5, 0.7, 0.9$. Median reversal magnitudes range from

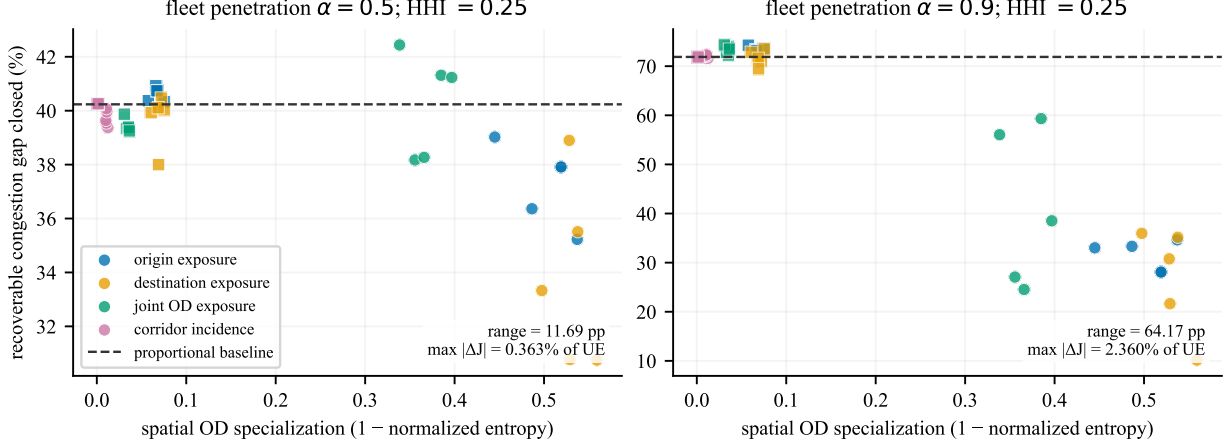


Figure 7: Spatial OD portfolios change congestion at fixed equal fleet masses and HHI. The figure is a complementary exposure test, not a market-population estimate.

0.006% to 0.043% of J^{UE} . More HDVs can absorb a larger share of coalition displacement, reducing practical effect sizes; this transmission mechanism does not impose an ownership or coalition ordering. The earlier 200-profile decomposition scan remains a robustness check: HHI–recovery Spearman correlations are 0.967, 0.975, and 0.988 under the mass-linked prior, full-information $\beta = 1$, and common $\beta = 0.5$ regimes, with all-pair reversal rates 7.97%, 7.05%, and 4.14%. These proportional-portfolio results are secondary to the regime and policy analysis.

6.12 Identity-preserving coordination versus operational pooling

Pooling is a separate feasible-authority intervention, not another label for objective coordination. We compare the same two-coalition objective, company masses, aggregate OD demand, and $\gamma = 1$ under identity-preserving and operational-pooling constraints. With proportional member OD portfolios and common route access, pooling changes member labels but not the induced aggregate route-flow set, as established in Proposition 1. Four certified profiles confirm the boundary: the largest pooling-minus-identity TSTT difference is 0.003 time units, or $3.94 \times 10^{-10} J^{UE}$, and the largest recovery difference is 1.0308357×10^{-6} percentage points. Pooling is neutral in this symmetric design. The result is a scope boundary, not evidence that pooling is neutral with heterogeneous route access or spatially specialized member portfolios.

6.13 Summary of the evidence

The experiments support a regime and policy conclusion rather than a universal concentration law. Objective composition changes the coalition response only when cross-member terms are active; OD portfolios and route geometry determine where that response points; and HDV support determines how much reaches aggregate congestion. The analytic bottleneck threshold explains why either centralized or partial governance can be preferred. The cross-network scans show the same broad transition from grand to partial coordination as AV penetration and governance intensity rise, although the boundary is topology-specific. A local one-step merge rule has low but nonzero regret, and two-step lookahead reaches the audit optimum in all tested states. Governance costs then generate explicit grand, partial, and fragmented intervals. The result is a falsifiable selection

framework for specified arrangements, not a claim that firms will voluntarily form the selected coalition.

7 Discussion: From Coalition Effects to Coalition Policy

7.1 The organizational question resolved by the model

The paper begins with a routing-control question rather than a concentration question. AV companies face a common physical congestion technology but can use different management objectives for their controlled vehicles. Coalition governance determines which of those objectives are optimized jointly and which cross-member effects are internalized. Because each member retains its accepted OD demand, an organizational change first alters an OD-feasible route response. The road network maps that response into edge flows, and HDVs then absorb the component they can reproduce through Wardrop reallocation. Only the remaining activated response can change aggregate congestion.

This mechanism explains why coalition degree lacks a universal ordering. A larger coalition need not generate a larger or more socially aligned activated response. Its cross-member terms can rotate the response toward a different set of congested edges, and HDV substitution can remove some directions but not others. The relevant policy object is therefore the system-cost projection of the screened response. Proposition 8 makes this statement operational: a proposed increase in coupling improves TSTT when its merge-direction score is negative, is neutral when the score is zero, and is harmful when the score is positive.

7.2 When centralization and partial independence are preferred

The canonical theorem turns the general directional criterion into a regime map. When interior HDVs can offset the coalition response, they pin the congested route at the Wardrop flow and organization is locally neutral. Once that screening support disappears, the fixed component H and effective coalition response Σ determine system performance. If $H \geq x^{UE}/2$, TSTT is nondecreasing in Σ , so a centralizing move is desirable when it reduces effective response. If $H < x^{UE}/2$, system cost is minimized at the interior target $\Sigma^* = 1 - 2H/x^{UE}$. A partial coalition can then dominate both complete centralization and complete fragmentation because it produces a response closer to the target.

The theorem does not assert that a merger always reduces Σ . That relationship depends on the declared objective family and governance matrix. It instead states a sharper condition: once the response induced by an organization is known or estimated, the preferred direction follows from its position relative to the active regime. On a general network, the scalar target becomes the merge-direction score, which must be recomputed after a support change.

7.3 What the network evidence adds

The Sioux Falls audit scan is consistent with the regime logic. At $\alpha = 0.5$, the grand coalition minimizes TSTT for both tested positive governance intensities. At $\alpha = 0.9$, the optimal set retains independent decision centers: symmetry-equivalent three-one partitions at $\gamma = 0.5$ and a two-one-one partition at $\gamma = 1$. The two-network validation shows the same qualitative transition. All tested Braess and grid states with $\alpha \leq 0.7$ select the grand coalition, whereas high penetration and sufficiently strong governance produce partial optima on both networks. The transition occurs at different values of γ , so topology changes the boundary without eliminating the regime.

These results improve upon a list of partition rankings in two ways. First, the analytic theorem explains why an interior organization can be optimal. Second, policy regret asks whether a decision rule can recover the organization without treating the full partition lattice as the prescription. One-step merging is exact in all four Sioux Falls states and 29 of 32 cross-network states, with only 0.332 points of mean cross-network regret. Its three Braess failures are informative: local improvement can stop before a jointly valuable sequence of merges becomes visible. Two-step lookahead crosses these local traps and reaches an audit optimum in all 36 states. This observed exactness does not establish universal optimality, particularly because the four-company two-step neighborhood covers much of the small partition lattice.

7.4 Governance burden changes the recommendation

Minimizing TSTT alone treats coordination as costless. The burden model in (43) separates the network benefit from the resources needed to maintain cross-company governance. Each arrangement then defines an affine loss in λ : its congestion performance is the intercept and its coordination burden is the slope. The lower envelope yields exact switching thresholds. In the Sioux Falls example at $\alpha = 0.5, \gamma = 1$, the selected organization moves from grand coordination to two pairs, then to one pair, and finally to singletons as the governance-cost weight rises.

This result gives partial coalitions a substantive interpretation. They are not arbitrary intermediate points or evidence that moderation is always best. They appear when the marginal congestion benefit of an additional cross-member link no longer covers its governance burden. Alternative cost functions could change the thresholds or remove an intermediate interval; the lower-envelope method remains valid once that burden is specified.

7.5 Why HHI cannot implement the policy

The response-based policy needs variables that a passive ownership cross section does not reveal. The controlled composition experiment holds company and coalition HHI, masses, coalition sizes, OD obligations, and route access fixed, yet changing which objective types are grouped alters recovery and can reverse the ordering along the governance path. Across the full Sioux Falls audit, the coalition-HHI/recovery Spearman correlation falls to 0.236 and the pairwise inversion rate reaches 40.74% at high penetration and full governance. At fixed company HHI, changing spatial OD portfolios creates even larger variation because it changes the network directions controlled by each company.

HHI remains useful for describing the distribution of fleet mass. It cannot select a coalition organization because it omits objective curvature, cross-member governance, OD authority, route access, active support, and HDV screening. The policy should therefore treat HHI as a screening covariate and estimate or simulate the activated response directly.

7.6 Observability and implementation

Implementation requires four classes of inputs: accepted company-level OD obligations, available routes or dispatch authority, calibrated physical delay functions, and a defensible approximation of each company’s marginal management objective. Under a stable support, OD-preserving perturbations can identify screened aggregate-flow responses and provide the ingredients for the merge score. When supports change, the manager can use the certified continuation solver to evaluate the local one- or two-step merge neighborhood. This creates an auditable workflow: diagnose the response regime, evaluate candidate local merges, report regret against richer audit scans when feasible, and include governance burden before recommending an organization.

The recommendation applies to an operator, platform, or regulator that can compare specified governance arrangements. It does not imply that every member would accept the system-optimal continuation design. Private implementation requires a payoff-allocation or transfer rule, contracting costs, deviation protocols, and a partition-formation concept.

7.7 Scope and limitations

The exact response and regime results require affine physical costs, fixed active supports, nonredundant OD tangents, and positive curvature on coalition-feasible directions. The BPR experiments provide certified numerical continuation equilibria, not a global nonlinear extension of the affine theorem. The governance burden is a transparent sensitivity primitive rather than an externally calibrated institutional cost. Objective types are also controlled primitives; estimating them from operator dispatch data remains an important empirical task.

The routing stage conditions on an externally realized batch of accepted requests. Pricing, passenger choice, endogenous demand, empty-vehicle repositioning, service quality, capacity and headway effects, private profit, and endogenous coalition stability remain outside the model. The complete partition scans cover four-company benchmark menus rather than a population of markets, and the two-step policy has no general optimality guarantee. These boundaries leave a clear next step: combine empirically identified objectives with scalable local search and a partition-function formation game, while retaining the continuation-equilibrium mechanism established here.

8 Conclusion

This paper asks when heterogeneous AV companies should coordinate their routing and when partial independence should be retained. Companies experience the same physical road costs but can manage their fleets under different objectives. A coalition aggregates those objectives and internalizes selected cross-member effects while preserving member OD obligations; HDVs continue to choose routes through a nonatomic Wardrop equilibrium. The resulting organizational problem concerns the continuation congestion induced by a specified coalition, not the ownership shares alone and not the private stability of coalition formation.

The theory connects objective heterogeneity to a usable coalition-design criterion. On a fixed affine support, the activated signature retains the OD-feasible coalition response that survives HDV reallocation. Its projection onto marginal system cost yields a general-network merge-direction score. The canonical bottleneck exposes three regimes: HDV screening can make organization neutral; a boundary regime favors changes that lower effective response; and an interior-response regime can make partial coordination better than both grand and singleton organization. Adding governance burden gives closed-form pairwise thresholds and a lower envelope of grand, partial, and fragmented policy intervals.

The certified experiments test both the regime and the selection rule. Sioux Falls moves from a grand-coalition optimum at moderate penetration to partial coordination at high penetration. Independent Braess and directed-grid scans show the same qualitative transition, with topology-specific boundaries. A one-step merge policy reaches an audit optimum in 33 of 36 total states and has mean regret 0.332 points on the 32 cross-network states; two-step lookahead reaches an audit optimum in all 36. The latter result is evidence within the declared four-company designs rather than a universal guarantee. Objective-composition, company-count, spatial-exposure, and sensitivity tests explain why the policy cannot be replaced by HHI or coalition count.

The practical conclusion is conditional but specific. A decision maker should estimate or simulate the activated response, use its system-cost projection to screen candidate merges, search a

local merge neighborhood, and account for the cost of cross-company governance. Centralization is justified when that process moves the network toward its response target and the congestion gain covers the coordination burden. Partial independence is justified when further coupling overshoots the target or costs more than it saves. Extending this continuation design to empirically calibrated objectives and endogenous coalition formation is a separate research step.

Data Availability Statement

All analytical derivations, experiment scripts, machine-readable result files, figure generators, and numerical validators used for this manuscript are included in the repository accompanying the paper. The Sioux Falls network inputs are identified in [Appendix B](#).

Ethics Statement

This study uses public benchmark network data and controlled computational scenarios. It does not involve human participants, personal data, or animal subjects.

Author Contributions

Luyao Niu: conceptualization, methodology, mathematical analysis, software, experiments, and writing. Ruolin Li: conceptualization, methodology, mathematical analysis, review, and writing.

Conflict of Interest

The authors declare no conflict of interest.

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Generative AI tools were used for drafting assistance, code-review support, and language editing. The authors verified the mathematical derivations, experimental outputs, citations, and final manuscript and remain responsible for all content.

A Additional Analytical Details

A.1 Quadratic coalition objectives and private travel time

This subsection clarifies the economic normalization behind the quadratic specification. The integral in (8) supplies the common marginal road cost. A quadratic term is then needed if the coalition is meant to internalize the delay generated by its own flow rather than merely take that delay as given.

For affine costs, the physical part of coalition C 's objective on edge e is

$$\int_0^{F_e^C} [a_e + b_e(x_e^{-C} + s)] ds = a_e F_e^C + b_e x_e^{-C} F_e^C + \frac{1}{2} b_e (F_e^C)^2.$$

Coalition travel time on that edge is

$$F_e^C [a_e + b_e(x_e^{-C} + F_e^C)] = a_e F_e^C + b_e x_e^{-C} F_e^C + b_e (F_e^C)^2.$$

Thus the management penalty $\frac{1}{2} b_e (F_e^C)^2$ converts the physical integral into the coalition's private total travel time on that edge. The equality is a normalization for the affine case, not a claim that every application must use total travel time as its private objective. For a singleton, $F_e^C = f_e^i$. For a fully integrated identity-preserving coalition, the governance matrix $b_e \mathbf{1}_C \mathbf{1}_C^\top$ produces exactly this aggregate square. A diagonal matrix retains separate member objectives; off-diagonal elements add cross-member internalization. The general quadratic form in (10) allows these terms to coexist with route-specific priorities q_C .

A.2 Projected inverse in a null-space basis

Let Z_C have orthonormal columns spanning $\ker \Gamma_C$. Any feasible active flow is $y_C^0 + Z_C u$. The coalition's fixed-support quadratic first-order condition restricted to this tangent is

$$(Z_C^\top H_C Z_C) u = -Z_C^\top [H_C y_C^0 + \Lambda_C^\top (a + Bx) + q_C].$$

The tangent positive-definiteness condition (19) makes the coefficient invertible and gives

$$P_C = Z_C (Z_C^\top H_C Z_C)^{-1} Z_C^\top.$$

This formula remains valid when H_C is singular on route-space directions that cannot be generated by an OD-preserving movement. Exact duplicate routes must be combined or quotiented out when they create a zero-curvature vector in $\ker \Gamma_C$. The condition is therefore a statement about curvature on feasible deviations, not a requirement that the primitive objective be strictly convex in every route coordinate.

A.3 HDV projection identities

The HDV block enters the theorem through feasible reallocation directions, not through an additional management objective. The matrix $\mathcal{A}_H$ removes the edge-flow component that HDVs can generate while preserving their OD totals and equal-cost condition.

Let $K_H = L_H^\top B L_H$. Under (27),

$$\mathcal{A}_H = I - L_H K_H^{-1} L_H^\top B.$$

The identities in Lemma 2 follow from

$$\mathcal{A}_H L_H = L_H - L_H K_H^{-1} K_H = 0$$

and

$$\mathcal{A}_H^2 = I - 2L_H K_H^{-1} L_H^\top B + L_H K_H^{-1} K_H K_H^{-1} L_H^\top B = \mathcal{A}_H.$$

If $\mathcal{A}_H v = 0$, then $v = L_H K_H^{-1} L_H^\top B v$, proving $\ker \mathcal{A}_H = \text{range}(L_H)$. Symmetry of B and K_H gives $\mathcal{A}_H^\top B = B \mathcal{A}_H$.

A.4 BPR continuation diagnostic

The exact activated-response theorem is affine. It should not be read as a closed-form solution for the BPR continuation game. For BPR costs, the monotonicity proof in Proposition 4 applies directly because BPR costs are continuous and nondecreasing and the coalition management matrices are PSD. The earlier perceived-slope specification has a more restrictive pointwise symmetric-Jacobian condition. On one edge, with fleet flows f^k , coefficients β_k , and $x = \sum_k f^k + h$, the condition is

$$\frac{c''(x)}{c'(x)} \sqrt{\sum_k \beta_k (f^k)^2} \leq 2. \quad (49)$$

For a BPR exponent $\eta > 1$, a readily checked sufficient bound is

$$\beta_{\max} \sigma \leq \frac{4}{(\eta - 1)^2}, \quad \sigma = \max_k f^k / x. \quad (50)$$

At $\eta = 4$ the threshold is $4/9$. A pointwise pass records membership in a sufficient region; a fail removes that guarantee but is not evidence of multiplicity. All BPR profiles in the paper are certified independently by VI residual and omitted-route tests. Those numerical certificates establish approximate equilibrium of the solved route-generation problem; they do not extend the affine signature theorem to nonlinear BPR costs.

A.5 Symbolic check of the canonical regime derivative

For the bottleneck theorem, substituting $x_1 = (H + \Sigma x^{UE}) / (1 + \Sigma)$ into $J = Na + t_1(x_1^2 - x^{UE}x_1)$ and differentiating gives

$$\frac{dJ}{d\Sigma} = t_1(2x_1 - x^{UE}) \frac{x^{UE} - H}{(1 + \Sigma)^2}.$$

The identity

$$2x_1 - x^{UE} = \frac{2H + (\Sigma - 1)x^{UE}}{1 + \Sigma}$$

then yields (39). The repository script `verify_regime_theorem.py` checks this simplification symbolically and records the zero residual and threshold in `regime_theorem_verification.json`. The script is a reproducibility check; the proof in Theorem 2 supplies the mathematical argument.

B Sioux Falls Data and Numerical Certification

Sioux Falls has 24 nodes, 76 directed links, and 528 positive-demand OD pairs. Demand, free-flow times, and capacities are read from the standard network files in the repository. Total demand is 360,600. Physical costs use (3). The all-HDV user-equilibrium and system-optimum TSTTs are 7,480,226 and 7,194,256, respectively.

The initial route set contains the three shortest free-flow routes per OD. It is expanded by player-specific shortest-path column generation, because a route that is unattractive at free flow can become attractive under the marginal objective induced by coalition governance. A route is added when its current marginal route cost is below the minimum held-route marginal cost by more than 10^{-6} times the route-cost scale. The VI is solved with adaptive Korpelovich extragradient and Euclidean projection onto each identity-preserving company-OD simplex. The accepted profiles satisfy normalized VI gap at most 10^{-6} and full-network omitted-route slack at most 10^{-6} .

The composition experiment contains 40 profiles for four equal-mass companies with objective types (0.5, 0.5, 1.5, 1.5). It crosses two penetration values, five coordination values, and four partitions. The degree experiment contains 54 profiles across $n \in \{2, 4, 8\}$, two penetration values, three coordination values, and all declared balanced partitions.

The sensitivity experiment contains 180 profiles crossing three objective gaps, three reference-slope scales, five coordination values, two penetrations, and the assortative/mixed partition pair. The complete $n = 4$ partition scan contains all 15 set partitions at two positive coordination values and two penetrations, for 60 profiles. The pooling decomposition contains four profiles comparing identity-preserving and operational-pooling feasibility. All 338 Sioux Falls coalition-extension profiles are certified.

The cross-network audit contains a five-edge Braess topology and a directed 3×3 grid with three positive-demand ODs. For each topology, the design crosses four AV penetration levels, four governance intensities, and all 15 partitions of four companies, for 480 profiles. All profiles are certified; the maximum normalized VI gap is 9.971×10^{-7} and the maximum relative omitted-route slack is 3.79×10^{-16} . The raw profile file is `cross_network_regime_results.json`; the state and policy summaries are in `cross_network_regime_summary.json`.

Policy regret is computed after, rather than during, equilibrium solution. `coalition_design_analysis.py` calculates set-valued Sioux Falls optima, one- and two-step merge paths, fixed-policy regret, and the governance-cost lower envelope. `cross_network_policy_analysis.py` applies the same local policies to the two independent topologies. The validator `validate_regime_package.py` checks profile counts, certification thresholds, finiteness, Sioux Falls policy consistency, and the reported cross-network regret values.

The fixed-HHI spatial experiment contains 82 certified profiles. Four equal-mass companies have HHI 0.25 at both $\alpha = 0.5$ and 0.9; iterative proportional fitting preserves every aggregate OD row and every fleet-mass column while varying spatial OD portfolios. The response-decomposition experiment contains 600 certified rows, the penetration scan 500, and the targeted BPR audit 10. The raw and summary JSON files, scripts, and validators are included in the repository.

Table 9: Numerical certification summary. All rows satisfy both acceptance criteria. Exact PSD entries are pointwise diagnostics for the earlier perceived-slope family, not acceptance criteria for the new coalition quadratic family.

Experiment	certified	max VI gap	max relative oracle slack	rows
coalition composition	40/40	1.0×10^{-6}	1.0×10^{-6}	40
coalition degree	54/54	9.97×10^{-7}	9.91×10^{-7}	54
coalition sensitivity	180/180	1.0×10^{-6}	1.0×10^{-6}	180
all $n = 4$ partitions	60/60	1.0×10^{-6}	1.0×10^{-6}	60
Braess/grid regime audit	480/480	9.97×10^{-7}	3.79×10^{-16}	480
identity/pooling decomposition	4/4	9.84×10^{-7}	3.2×10^{-8}	4
fixed-HHI spatial OD	82/82	1.0×10^{-6}	1.0×10^{-6}	82
response decomposition	600/600	1.0×10^{-6}	9.0×10^{-7}	600
penetration scan	500/500	1.0×10^{-6}	4.2×10^{-7}	500
BPR anchor audit	10/10	1.0×10^{-6}	1.2×10^{-7}	10

C Reproducibility and Scope Ledger

The current coalition solver extends the previous fleet solver through optional coalition governance arguments while preserving the earlier model as a default nested case. The coalition objective uses fixed reference curvature $\bar{b}_e = c'_e(x_e^{UE})$ and the PSD matrix in (13). The company-count scan uses equal company masses within each n and holds total AV mass fixed across n . Singleton profiles

at nonzero γ are generated from the analytically verified partition invariance at $\gamma = 0$; they are labeled as derived rows in the raw JSON and are not counted as independent solver runs. These implementation choices define the declared counterfactual menu; they do not estimate how firms would choose a coalition in an unfixed market.

The empirical claims in Section 6 are limited to the declared Sioux Falls benchmark and governance design. The objective types are controlled primitives, not estimates from operator data. Operational pooling, endogenous coalition stability, transfers, pricing, passenger choice, and platform welfare remain outside the reported core results. The theoretical claims are likewise conditional: existence and monotonicity use continuous nondecreasing physical costs, whereas the activated signature and its minimality use affine costs, fixed supports, and the stated tangent-rank conditions.

The repository includes the extension and cross-network scripts, raw and summary JSON outputs, symbolic checks, fail-closed validators, and figure-generation code. Pooling is implemented as a coalition-level simplex over member-route coordinates; its comparison with identity-preserving coordination is interpreted only under the proportional OD and common-route design reported in Section 6. The local merge policies use full scans only as audit baselines for regret. Their zero observed two-step regret is not encoded as an assumption and is not claimed as a general optimality theorem.

References